Rajasthan Board

Session 2026–27

CLASS XII

SENIOR SECONDARY EXAMINATION

COMPUTER SCIENCE (Optional)

1000 QUESTIONS & ANSWERS

COMPLETE QUESTION BANK
WITH DETAILED SOLUTIONS

- Based on Latest Syllabus
- Previous Year Papers (2024–2026)
- Chapter-wise Coverage
- MCQs, VSA, SA, LA Questions
- Programming & SQL Questions
- Application-based & Competency Questions
- Examiner's Trap Zone
- Quick Revision Notes



COVERS ALL 12 LESSONS

01	Exception Handling	07	Understanding Data
02	File Handling	08	Database Concepts
03	Stack	09	Structured Query Language (SQL)
04	Queue	10	Computer Networks
05	Sorting	11	Data Communication
06	Searching	12	Security Aspects



1000 QUESTIONS WITH ANSWERS ★★★

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CLASS XII COMPUTER SCIENCE (Optional) 1000 QUESTIONS & ANSWERS Session 2026–27

LESSON 01 — EXCEPTION HANDLING

A. Multiple Choice Questions

1. Which error occurs when Python code violates syntax rules?

A) SyntaxError B) ValueError C) TypeError D) NameError

Answer: (A)

2. Which keyword explicitly raises an exception?

A) try B) raise C) except D) finally

Answer: (B)

3. Which exception occurs when a number is divided by zero?

A) ValueError B) TypeError C) ZeroDivisionError D) IndexError

Answer: (C)

4. Which block contains code that may generate an exception?

A) try B) except C) else D) finally

Answer: (A)

5. Which block handles a matching exception?

A) try B) except C) else D) raise

Answer: (B)

6. When does the else block normally execute?

A) Always B) Only if try raises C) If try completes without an exception D) Never

Answer: (C)

7. Which block is intended for cleanup code?

A) else B) except C) finally D) raise

Answer: (C)

8. An undefined variable name commonly causes:

A) NameError B) IndexError C) EOFError D) KeyError

Answer: (A)

9. An inappropriate operand type commonly causes:

A) TypeError B) NameError C) SyntaxError D) IOError

Answer: (A)

10. assert is mainly used to:

A) Open a file B) Validate a condition C) Sort a list D) Define a class

Answer: (B)

B. Fill in the Blanks

1. An error caused by violating language syntax rules is called a _____.

Answer: SyntaxError

2. The keyword used to explicitly raise an exception is _____.

Answer: raise

3. The block that executes whether or not an exception occurs is _____.

Answer: finally

4. An exception caused by division by zero is _____.

Answer: ZeroDivisionError

5. A Python object representing an error during execution is called an _____.

Answer: exception

C. True / False

1. A syntactically correct program can still raise an exception during execution.

Answer: True

2. The finally block executes only when an exception occurs.

Answer: False

3. Multiple except clauses may be associated with one try block.

Answer: True

4. NameError may occur when a variable is not defined.

Answer: True

5. raise and assert are identical in purpose.

Answer: False

D. Very Short Answer Questions

1. What is an exception?

Answer: An exception is a Python object representing an error or abnormal condition that occurs during program execution.

2. What is a syntax error?

Answer: It is an error caused by not following the syntax rules of Python.

3. What is the purpose of try?

Answer: The try block contains statements in which an exception may occur.

4. What is the purpose of except?

Answer: The except block contains code that handles a matching exception.

5. What is the purpose of finally?

Answer: It contains cleanup or essential statements that should execute whether or not an exception occurs.

E. Short Answer Questions

1. Differentiate between a syntax error and an exception.

Answer: A syntax error prevents the program from being parsed/executed because the code violates Python syntax. An exception occurs during execution of otherwise syntactically valid code, such as division by zero.

2. Explain raise with an example.

Answer: raise explicitly generates an exception. Example: raise ValueError("Invalid marks"). It transfers control to an appropriate handler if one is available.

3. Explain try-except.

Answer: Statements that may fail are placed in try. If a matching exception occurs, control moves to the corresponding except block, allowing the program to handle the error instead of terminating abruptly.

4. Differentiate raise and assert.

Answer: raise is used to explicitly raise a specified exception. assert checks a condition and raises AssertionError when the condition is false.

5. Why is exception handling useful?

Answer: It prevents uncontrolled termination, separates normal logic from error-handling logic, and allows meaningful recovery or messages.

F. Long Answer / Programming Questions

1. Explain try-except-else-finally with a suitable flow.

2. Write a program to handle invalid input and division by zero.

3. Explain built-in and user-defined exceptions.

4. Explain traceback and exception propagation.

5. Explain multiple except clauses.

Additional Questions

1. [MCQ] Which of the following is associated with the error raised when a variable name is used before it is defined?

A) NameError B) TypeError C) IndexError D) ValueError

Answer: NameError

Explanation: NameError indicates that a local or global name could not be found.

2. [MCQ] Which of the following is associated with the error raised when an operation receives an inappropriate data type?

A) TypeError B) ValueError C) NameError D) KeyError

Answer: TypeError

Explanation: TypeError concerns an inappropriate type of operand or argument.

3. [MCQ] Which of the following is associated with the error raised when a valid type has an inappropriate value?

A) ValueError B) TypeError C) NameError D) IndexError

Answer: ValueError

Explanation: ValueError occurs when the type is acceptable but the value is unsuitable.

4. [MCQ] Which of the following is associated with the error raised when a sequence index is outside its valid range?

A) IndexError B) KeyError C) NameError D) ValueError

Answer: IndexError

Explanation: IndexError is used for an invalid sequence index.

5. [MCQ] Which of the following is associated with the exception raised when a dictionary key is not found?

A) KeyError B) IndexError C) NameError D) TypeError

Answer: KeyError

Explanation: KeyError occurs for a missing mapping key.

6. [MCQ] Which of the following is associated with the keyword used to explicitly raise an exception?

A) raise B) assert C) throw D) except

Answer: raise

Explanation: The raise statement explicitly triggers an exception.

7. [MCQ] Which of the following is associated with the statement used to test an assumption and raise AssertionError if it is false?

A) assert B) raise C) check D) verify

Answer: assert

Explanation: assert evaluates a condition and raises AssertionError when false.

8. [MCQ] Which of the following is associated with the block containing statements that may cause an exception?

A) try B) except C) else D) finally

Answer: try

Explanation: Risky statements are placed in the try block.

9. [MCQ] Which of the following is associated with the block used to handle a matching exception?

A) except B) try C) else D) finally

Answer: except

Explanation: An except clause contains exception-handling code.

10. [MCQ] Which of the following is associated with the optional block that runs when the try block completes without an exception?

A) else B) except C) finally D) raise

Answer: else

Explanation: The else clause runs only when no exception occurs in try.

11. [MCQ] Which of the following is associated with the block normally used for cleanup operations?

A) finally B) else C) except D) try

Answer: finally

Explanation: finally is intended for code that should run regardless of normal exception handling.

12. [MCQ] Which of the following is associated with the report showing the sequence of calls leading to an exception?

A) traceback B) syntax tree C) file handle D) data dictionary

Answer: traceback

Explanation: A traceback shows the call sequence and location associated with an exception.

13. [MCQ] Which of the following is associated with the mechanism that separates normal program logic from error-handling code?

A) exception handling B) sorting C) serialization D) indexing

Answer: exception handling

Explanation: Exception handling manages abnormal execution without mixing all error logic into normal logic.

14. [MCQ] Which of the following is associated with the error caused by dividing a number by zero?

A) ZeroDivisionError B) ArithmeticError only C) ValueError D) OverflowError

Answer: ZeroDivisionError

Explanation: Python raises ZeroDivisionError for division or modulo by zero.

15. [MCQ] Which of the following is associated with the error associated with incorrect indentation?

A) IndentationError B) IndexError C) TypeError D) KeyError

Answer: IndentationError

Explanation: IndentationError indicates invalid indentation in Python code.

16. [MCQ] Which of the following is associated with the exception associated with importing an unavailable module or name?

A) ImportError B) NameError C) FileNotFoundError D) EOFError

Answer: ImportError

Explanation: ImportError is raised when an import operation cannot be completed.

17. [MCQ] Which of the following is associated with the error raised when a requested file cannot be found?

A) FileNotFoundError B) NameError C) ValueError D) TypeError

Answer: FileNotFoundError

Explanation: FileNotFoundError is a file-system related exception.

1. [Fill] Complete the statement: the block containing statements that may cause an exception is _____.

Answer: try

Explanation: Risky statements are placed in the try block.

2. [Fill] Complete the statement: the block used to handle a matching exception is _____.

Answer: except

Explanation: An except clause contains exception-handling code.

3. [Fill] Complete the statement: the optional block that runs when the try block completes without an exception is _____.

Answer: else

Explanation: The else clause runs only when no exception occurs in try.

4. [Fill] Complete the statement: the block normally used for cleanup operations is _____.

Answer: finally

Explanation: finally is intended for code that should run regardless of normal exception handling.

5. [Fill] Complete the statement: the report showing the sequence of calls leading to an exception is _____.

Answer: traceback

Explanation: A traceback shows the call sequence and location associated with an exception.

6. [Fill] Complete the statement: the mechanism that separates normal program logic from error-handling code is _____.

Answer: exception handling

Explanation: Exception handling manages abnormal execution without mixing all error logic into normal logic.

7. [Fill] Complete the statement: the error caused by dividing a number by zero is _____.

Answer: ZeroDivisionError

Explanation: Python raises ZeroDivisionError for division or modulo by zero.

8. [Fill] Complete the statement: the error associated with incorrect indentation is _____.

Answer: IndentationError

Explanation: IndentationError indicates invalid indentation in Python code.

9. [Fill] Complete the statement: the exception associated with importing an unavailable module or name is _____.

Answer: ImportError

Explanation: ImportError is raised when an import operation cannot be completed.

1. [T/F] True or False: the error caused by dividing a number by zero is ArithmeticError only.

Answer: False

Explanation: Correct answer: ZeroDivisionError. Python raises ZeroDivisionError for division or modulo by zero.

2. [T/F] True or False: the error associated with incorrect indentation is IndentationError.

Answer: True

Explanation: IndentationError indicates invalid indentation in Python code.

3. [T/F] True or False: the exception associated with importing an unavailable module or name is NameError.

Answer: False

Explanation: Correct answer: ImportError. ImportError is raised when an import operation cannot be completed.

4. [T/F] True or False: the error raised when a requested file cannot be found is FileNotFoundError.

Answer: True

Explanation: FileNotFoundError is a file-system related exception.

5. [T/F] True or False: a programmer-created exception class is built-in exception.

Answer: False

Explanation: Correct answer: user-defined exception. A user-defined exception is normally created by defining a class derived from Exception.

6. [T/F] True or False: the purpose of multiple except clauses is handle different exception types separately.

Answer: True

Explanation: Multiple except clauses allow different exception types to receive different handlers.

7. [T/F] True or False: the normal control flow after an exception is handled by an except clause is the try block restarts automatically.

Answer: False

Explanation: Correct answer: the handler executes and the try block is abandoned. After an exception is raised, remaining statements in that try block are skipped.

1. [VSA] Name the concept associated with: the error raised when a sequence index is outside its valid range.

Answer: IndexError

Explanation: IndexError is used for an invalid sequence index.

2. [VSA] Name the concept associated with: the exception raised when a dictionary key is not found.

Answer: KeyError

Explanation: KeyError occurs for a missing mapping key.

3. [VSA] Name the concept associated with: the keyword used to explicitly raise an exception.

Answer: raise

Explanation: The raise statement explicitly triggers an exception.

4. [VSA] Name the concept associated with: the statement used to test an assumption and raise AssertionError if it is false.

Answer: assert

Explanation: assert evaluates a condition and raises AssertionError when false.

5. [VSA] Name the concept associated with: the block containing statements that may cause an exception.

Answer: try

Explanation: Risky statements are placed in the try block.

6. [VSA] Name the concept associated with: the block used to handle a matching exception.

Answer: except

Explanation: An except clause contains exception-handling code.

7. [VSA] Name the concept associated with: the optional block that runs when the try block completes without an exception.

Answer: else

Explanation: The else clause runs only when no exception occurs in try.

8. [VSA] Name the concept associated with: the block normally used for cleanup operations.

Answer: finally

Explanation: finally is intended for code that should run regardless of normal exception handling.

1. [Short/Application] Explain the following concept: the keyword used to explicitly raise an exception.

Answer: raise

Explanation: The raise statement explicitly triggers an exception.

2. [Short/Application] Explain the following concept: the statement used to test an assumption and raise AssertionError if it is false.

Answer: assert

Explanation: assert evaluates a condition and raises AssertionError when false.

3. [Short/Application] Explain the following concept: the block containing statements that may cause an exception.

Answer: try

Explanation: Risky statements are placed in the try block.

4. [Short/Application] Explain the following concept: the block used to handle a matching exception.

Answer: except

Explanation: An except clause contains exception-handling code.

5. [Short/Application] Explain the following concept: the optional block that runs when the try block completes without an exception.

Answer: else

Explanation: The else clause runs only when no exception occurs in try.

6. [Short/Application] Explain the following concept: the block normally used for cleanup operations.

Answer: finally

Explanation: finally is intended for code that should run regardless of normal exception handling.

7. [Short/Application] Explain the following concept: the report showing the sequence of

calls leading to an exception.

Answer: traceback

Explanation: A traceback shows the call sequence and location associated with an exception.

8. [Short/Application] Explain the following concept: the mechanism that separates normal program logic from error-handling code.

Answer: exception handling

9. [Short/Application] Explain the following concept in detail: the error caused by dividing a number by zero.

Answer: ZeroDivisionError

Explanation: Python raises ZeroDivisionError for division or modulo by zero.

LESSON 02 — FILE HANDLING

A. Multiple Choice Questions

1. Which function opens a file in Python?

A) open() B) start() C) fileopen() D) access()

Answer: (A)

2. The object returned by open() is commonly called a:

A) File handle B) Loop handle C) String handle D) Record key

Answer: (A)

3. Default mode of open() is:

A) w B) a C) r D) x

Answer: (C)

4. Which mode appends data at the end?

A) r B) w C) a D) x

Answer: (C)

5. Which method writes one string?

A) read() B) write() C) readline() D) dump()

Answer: (B)

6. Which method returns all lines as a list?

A) readlines() B) readline() C) read() D) tell()

Answer: (A)

7. Which method returns the current file position?

A) seek() B) tell() C) close() D) position()

Answer: (B)

8. Which method moves the file pointer?

A) seek() B) tell() C) move() D) shift()

Answer: (A)

9. Which module is used for pickling?

A) pickle B) binary C) object D) files

Answer: (A)

10. Which pickle function writes an object?

A) load() B) dump() C) write() D) store()

Answer: (B)

B. Fill in the Blanks

1. A file provides permanent storage for _____.

Answer: data

2. The default file mode is _____.

Answer: r

3. The method used to close a file is _____.

Answer: close()

4. The process of converting an object into a byte stream is called _____.

Answer: serialization/pickling

5. pickle._____() writes an object to a binary file.

Answer: dump

C. True / False

1. A binary file is normally human-readable in a text editor.

Answer: False

2. w mode can overwrite existing contents.

Answer: True

3. The with clause can automatically close a file.

Answer: True

4. writelines() automatically adds a newline after every string.

Answer: False

5. tell() changes the file position.

Answer: False

D. Very Short Answer Questions

1. Name the two main types of data files.

Answer: Text files and binary files.

2. What is a file handle?

Answer: It is the file object returned by open(), linking the program with the file.

3. What does read() do?

Answer: It reads a specified amount of data, or the complete file when used without an argument.

4. What does readline() do?

Answer: It reads one line from a text file.

5. What are tell() and seek() used for?

Answer: tell() reports the current position and seek() moves the file object to a specified position.

E. Short Answer Questions

1. Differentiate text and binary files.

Answer: Text files store character data intended for human-readable representation. Binary files store data as bytes and are normally interpreted by specific software.

2. Differentiate write() and writelines().

Answer: write() writes a single string. writelines() writes a sequence/iterable of strings and does not automatically insert newline characters.

3. Explain r, w and a modes.

Answer: r reads an existing file, w opens for writing and may erase existing contents, while a opens for appending at the end and can create the file if necessary.

4. Why use with while handling files?

Answer: with manages the file context and closes the file automatically when control leaves the block, including after many exceptions.

5. Explain pickling and unpickling.

Answer: Pickling/serialization converts a Python object into a byte stream for storage. Unpickling/deserialization reconstructs the object from that stored representation.

F. Long Answer / Programming Questions

1. Explain the complete text-file handling cycle.

2. Write a Python program to copy a text file.

3. Explain binary-file record handling with pickle.

4. Differentiate read(), readline() and readlines().

5. Explain file pointer operations.

Additional Questions

1. [MCQ] Which of the following is associated with the function used to open a file?

A) open() B) file() C) start() D) access()

Answer: open()

Explanation: open() returns a file object.

2. [MCQ] Which of the following is associated with the default file mode?

A) r B) w C) a D) x

Answer: r

Explanation: The default mode of open() is read text mode, r.

3. [MCQ] Which of the following is associated with the mode that creates or truncates a text file for writing?

A) w B) r C) a D) x

Answer: w

Explanation: w opens for writing and truncates an existing file.

4. [MCQ] Which of the following is associated with the mode that appends data at the end of a file?

A) a B) w C) r D) x

Answer: a

Explanation: a opens for appending and positions writing at the end.

5. [MCQ] Which of the following is associated with the mode that creates a new file and fails if it already exists?

A) x B) a C) w D) r

Answer: x

Explanation: x is exclusive creation mode.

6. [MCQ] Which of the following is associated with the mode for reading and writing text data without truncating the file?

A) r+ B) w+ C) a+ D) rb+

Answer: r+

Explanation: r+ permits both reading and writing while requiring the file to exist.

7. [MCQ] Which of the following is associated with the method that reads the entire remaining file content as a string?

A) read() B) readline() C) readlines() D) write()

Answer: read()

Explanation: read() returns file content as a string, optionally limited by size.

8. [MCQ] Which of the following is associated with the method that reads one line?

A) readline() B) read() C) readlines() D) write()

Answer: readline()

Explanation: readline() reads a single line.

9. [MCQ] Which of the following is associated with the method that returns a list of lines?

A) readlines() B) read() C) readline() D) writelines()

Answer: readlines()

Explanation: readlines() returns a list containing the lines read.

10. [MCQ] Which of the following is associated with the method that writes a string to a text file?

A) write() B) read() C) append() D) put()

Answer: write()

Explanation: write() writes the supplied string and returns the number of characters written.

11. [MCQ] Which of the following is associated with the method that writes an iterable of strings?

A) writelines() B) writeall() C) readlines() D) append()

Answer: writelines()

Explanation: writelines() writes strings from an iterable; it does not add newlines automatically.

12. [MCQ] Which of the following is associated with the method that returns the current file position?

A) tell() B) seek() C) locate() D) position()

Answer: tell()

Explanation: tell() returns the current stream position.

13. [MCQ] Which of the following is associated with the method that changes the file position?

A) seek() B) tell() C) movefile() D) shift()

Answer: seek()

Explanation: seek() moves the file pointer to a specified position.

14. [MCQ] Which of the following is associated with the statement that automatically closes a file after its block?

A) with B) close C) finally only D) using

Answer: with

Explanation: A with statement manages the file context and closes it when the block ends.

15. [MCQ] Which of the following is associated with the Python module used for object serialization in the prescribed treatment?

A) pickle B) jsonfile C) binary D) osfile

Answer: pickle

Explanation: pickle supports serialization and deserialization of Python objects.

16. [MCQ] Which of the following is associated with the process of converting an object into a byte stream?

A) serialization/pickling B) deserialization C) indexing D) compilation

Answer: serialization/pickling

Explanation: Pickling serializes a Python object into a byte stream.

17. [MCQ] Which of the following is associated with the process of reconstructing an object from a byte stream?

A) deserialization/unpickling B) serialization C) parsing D) tokenization

Answer: deserialization/unpickling

Explanation: Unpickling reconstructs an object from serialized data.

18. [MCQ] Which of the following is associated with the binary mode used to read?

A) rb B) br C) rbin D) readb

Answer: rb

Explanation: rb means read binary.

19. [MCQ] Which of the following is associated with the binary mode used to write?

A) wb B) bw C) writeb D) bwrite

Answer: wb

Explanation: wb means write binary.

1. [Fill] Complete the statement: the method that reads one line is _____.

Answer: readline()

Explanation: readline() reads a single line.

2. [Fill] Complete the statement: the method that returns a list of lines is _____.

Answer: readlines()

Explanation: readlines() returns a list containing the lines read.

3. [Fill] Complete the statement: the method that writes a string to a text file is _____.

Answer: write()

Explanation: write() writes the supplied string and returns the number of characters written.

4. [Fill] Complete the statement: the method that writes an iterable of strings is _____.

Answer: writelines()

Explanation: writelines() writes strings from an iterable; it does not add newlines automatically.

5. [Fill] Complete the statement: the method that returns the current file position is _____.

Answer: tell()

Explanation: tell() returns the current stream position.

6. [Fill] Complete the statement: the method that changes the file position is _____.

Answer: seek()

Explanation: seek() moves the file pointer to a specified position.

7. [Fill] Complete the statement: the statement that automatically closes a file after its block is _____.

Answer: with

Explanation: A with statement manages the file context and closes it when the block ends.

8. [Fill] Complete the statement: the Python module used for object serialization in the prescribed treatment is _____.

Answer: pickle

Explanation: pickle supports serialization and deserialization of Python objects.

9. [Fill] Complete the statement: the process of converting an object into a byte stream is _____.

Answer: serialization/pickling

Explanation: Pickling serializes a Python object into a byte stream.

10. [Fill] Complete the statement: the process of reconstructing an object from a byte

stream is _____.

Answer: deserialization/unpickling

Explanation: Unpickling reconstructs an object from serialized data.

1. [T/F] True or False: the statement that automatically closes a file after its block is close.

Answer: False

Explanation: Correct answer: with. A with statement manages the file context and closes it when the block ends.

2. [T/F] True or False: the Python module used for object serialization in the prescribed treatment is pickle.

Answer: True

Explanation: pickle supports serialization and deserialization of Python objects.

3. [T/F] True or False: the process of converting an object into a byte stream is deserialization.

Answer: False

Explanation: Correct answer: serialization/pickling. Pickling serializes a Python object into a byte stream.

4. [T/F] True or False: the process of reconstructing an object from a byte stream is deserialization/unpickling.

Answer: True

Explanation: Unpickling reconstructs an object from serialized data.

5. [T/F] True or False: the binary mode used to read is br.

Answer: False

Explanation: Correct answer: rb. rb means read binary.

6. [T/F] True or False: the binary mode used to write is wb.

Answer: True

Explanation: wb means write binary.

7. [T/F] True or False: the common exception when a requested file does not exist is EOFError.

Answer: False

Explanation: Correct answer: FileNotFoundError. Opening a missing file for reading commonly raises FileNotFoundError.

8. [T/F] True or False: the purpose of close() is release the file resource and close the file.

Answer: True

Explanation: close() terminates the active file connection and releases associated resources.

1. [VSA] Name the concept associated with: the mode that appends data at the end of a file.

Answer: a

Explanation: a opens for appending and positions writing at the end.

2. [VSA] Name the concept associated with: the mode that creates a new file and fails if it already exists.

Answer: x

Explanation: x is exclusive creation mode.

3. [VSA] Name the concept associated with: the mode for reading and writing text data without truncating the file.

Answer: r+

Explanation: r+ permits both reading and writing while requiring the file to exist.

4. [VSA] Name the concept associated with: the method that reads the entire remaining file content as a string.

Answer: read()

Explanation: read() returns file content as a string, optionally limited by size.

5. [VSA] Name the concept associated with: the method that reads one line.

Answer: readline()

Explanation: readline() reads a single line.

6. [VSA] Name the concept associated with: the method that returns a list of lines.

Answer: readlines()

Explanation: readlines() returns a list containing the lines read.

7. [VSA] Name the concept associated with: the method that writes a string to a text file.

Answer: write()

Explanation: write() writes the supplied string and returns the number of characters written.

8. [VSA] Name the concept associated with: the method that writes an iterable of strings.

Answer: writelines()

Explanation: writelines() writes strings from an iterable; it does not add newlines automatically.

9. [VSA] Name the concept associated with: the method that returns the current file position.

Answer: tell()

Explanation: tell() returns the current stream position.

1. [Short/Application] Explain the following concept: the mode for reading and writing text data without truncating the file.

Answer: r+

Explanation: r+ permits both reading and writing while requiring the file to exist.

2. [Short/Application] Explain the following concept: the method that reads the entire remaining file content as a string.

Answer: read()

Explanation: read() returns file content as a string, optionally limited by size.

3. [Short/Application] Explain the following concept: the method that reads one line.

Answer: readline()

Explanation: readline() reads a single line.

4. [Short/Application] Explain the following concept: the method that returns a list of lines.

Answer: readlines()

Explanation: readlines() returns a list containing the lines read.

5. [Short/Application] Explain the following concept: the method that writes a string to a text file.

Answer: write()

Explanation: write() writes the supplied string and returns the number of characters written.

6. [Short/Application] Explain the following concept: the method that writes an iterable of strings.

Answer: writelines()

Explanation: writelines() writes strings from an iterable; it does not add newlines automatically.

7. [Short/Application] Explain the following concept: the method that returns the current file position.

Answer: tell()

8. [Short/Application] Explain the following concept in detail: the method that changes the file position.

Answer: seek()

Explanation: seek() moves the file pointer to a specified position.

9. [Short/Application] Explain the following concept in detail: the statement that automatically closes a file after its block.

Answer: with

Explanation: A with statement manages the file context and closes it when the block ends.

LESSON 03 — STACK

A. Multiple Choice Questions

1. A stack follows:

A) FIFO B) LIFO C) Random order D) Sorted order

Answer: (B)

2. Insertion in a stack is called:

A) pop B) push C) enqueue D) dequeue

Answer: (B)

3. Deletion from a stack is called:

A) push B) pop C) enqueue D) insert

Answer: (B)

4. The accessible end of a stack is called:

A) front B) rear C) top D) tail

Answer: (C)

5. Removing from an empty stack causes:

A) overflow B) underflow C) collision D) deadlock

Answer: (B)

6. Adding to a full fixed-size stack may cause:

A) underflow B) overflow C) EOF D) timeout

Answer: (B)

7. Which list method can implement push?

A) append() B) remove() C) sort() D) index()

Answer: (A)

8. Which notation places operators after operands?

A) Infix B) Prefix C) Postfix D) Middlefix

Answer: (C)

9. Which notation places operators before operands?

A) Prefix B) Infix C) Postfix D) Suffix

Answer: (A)

10. Which is a stack application?

A) Function call management B) IP addressing

C) File naming D) DNS lookup only

Answer: (A)

B. Fill in the Blanks

1. Stack follows the _____ principle.

Answer: LIFO

2. Insertion into a stack is called _____.

Answer: push

3. Deletion from a stack is called _____.

Answer: pop

4. The end from which stack operations occur is called the _____.

Answer: top

5. An attempt to pop an empty stack causes stack _____.

Answer: underflow

C. True / False

1. Stack follows FIFO.

Answer: False

2. append() can be used to push onto a Python-list stack.

Answer: True

3. pop() removes the top element.

Answer: True

4. A stack is primarily a linear data structure.

Answer: True

5. Function-call management can use a stack.

Answer: True

D. Very Short Answer Questions

1. What is a stack?

Answer: A stack is a linear data structure in which insertion and deletion take place at the same end, called the top.

2. What is LIFO?

Answer: LIFO means Last In, First Out.

3. What is peek?

Answer: Peek returns or views the top element without removing it.

4. What is stack overflow?

Answer: It is the condition in which an insertion is attempted when a fixed-size stack is full.

5. Name two applications of stack.

Answer: Function-call/recursion management and expression evaluation are two common applications.

E. Short Answer Questions

1. Explain push and pop.

Answer: Push inserts a new element at the top. Pop removes and returns the top element. Both are normally constant-time operations in a list-based stack.

2. Differentiate overflow and underflow.

Answer: Overflow occurs when insertion is attempted on a full fixed-size stack. Underflow occurs when deletion is attempted on an empty stack.

3. Differentiate stack and queue.

Answer: Stack follows LIFO and operates at one end. Queue follows FIFO, inserting at the rear and deleting from the front.

4. Explain stack in expression evaluation.

Answer: A stack stores operators or operands while an expression is converted or evaluated. Operators are applied when their precedence and position require it.

5. How is stack used in recursion?

Answer: Each function call creates an activation record on the call stack. When a call finishes, its record is removed, following LIFO order.

F. Long Answer / Programming Questions

1. Write a Python program for push, pop, peek and display.
2. Explain postfix evaluation using a stack.
3. Explain balanced-parentheses checking using a stack.
4. Explain stack applications.
5. Explain prefix, infix and postfix notations.

Additional Questions

1. [MCQ] Which of the following is associated with the access principle of a stack?

A) LIFO B) FIFO C) random access D) priority only

Answer: LIFO

Explanation: LIFO means Last In, First Out.

2. [MCQ] Which of the following is associated with the operation that adds an item to a stack?

A) push B) pop C) peek D) dequeue

Answer: push

Explanation: Push inserts an item at the top.

3. [MCQ] Which of the following is associated with the operation that removes the top item?

A) pop B) push C) peek D) enqueue

Answer: pop

Explanation: Pop removes the top item.

4. [MCQ] Which of the following is associated with the operation that reads the top item without removing it?

A) peek B) pop C) push D) displayall

Answer: peek

Explanation: Peek inspects the top element.

5. [MCQ] Which of the following is associated with the condition when an item is removed from an empty stack?

A) underflow B) overflow C) collision D) deadlock

Answer: underflow

Explanation: Underflow occurs when deletion is attempted on an empty stack.

6. [MCQ] Which of the following is associated with the condition when a stack cannot accept another item?

A) overflow B) underflow C) collision D) syntax error

Answer: overflow

Explanation: Overflow occurs when no space is available for another item in a bounded stack.

7. [MCQ] Which of the following is associated with the Python list method commonly used for push?

A) append() B) extend() C) remove() D) sort()

Answer: append()

Explanation: append() adds an item to the end, which can represent the top.

8. [MCQ] Which of the following is associated with the Python list method commonly used for pop?

A) pop() B) delete() C) remove all() D) cut()

Answer: pop()

Explanation: list.pop() removes and returns an item, commonly the top item.

9. [MCQ] Which of the following is associated with the notation with operators before operands?

A) prefix B) infix C) postfix D) suffix

Answer: prefix

Explanation: Prefix places the operator before its operands.

10. [MCQ] Which of the following is associated with the notation with operators between operands?

A) infix B) prefix C) postfix D) reverse

Answer: infix

Explanation: Infix is the usual human-readable arithmetic form.

11. [MCQ] Which of the following is associated with the notation with operators after operands?

A) postfix B) prefix C) infix D) middlefix

Answer: postfix

Explanation: Postfix is also called Reverse Polish notation.

12. [MCQ] Which of the following is associated with a common use of stack in expression processing?

A) parentheses matching B) packet routing C) database normalization

D) DNS resolution

Answer: parentheses matching

Explanation: Stacks can track opening parentheses and match them with closing parentheses.

13. [MCQ] Which of the following is associated with a common use of stack in program execution?

A) function-call management B) IP addressing

C) file naming D) SQL grouping

Answer: function-call management

Explanation: The call stack stores active function-call information.

14. [MCQ] Which of the following is associated with the data structure property of a

stack?

A) linear B) nonlinear only C) hierarchical D) relational

Answer: linear

Explanation: A stack is a linear data structure.

15. [MCQ] Which of the following is associated with the usual empty-stack test for a Python list named s?

A) len(s)==0 B) s==1 C) s.top()==0 D) empty(s)==False

Answer: len(s)==0

Explanation: An empty list has length zero.

1. [Fill] Complete the statement: the Python list method commonly used for pop is _____.

Answer: pop()

Explanation: list.pop() removes and returns an item, commonly the top item.

2. [Fill] Complete the statement: the notation with operators before operands is _____.

Answer: prefix

Explanation: Prefix places the operator before its operands.

3. [Fill] Complete the statement: the notation with operators between operands is _____.

Answer: infix

Explanation: Infix is the usual human-readable arithmetic form.

4. [Fill] Complete the statement: the notation with operators after operands is _____.

Answer: postfix

Explanation: Postfix is also called Reverse Polish notation.

5. [Fill] Complete the statement: a common use of stack in expression processing is _____.

Answer: parentheses matching

Explanation: Stacks can track opening parentheses and match them with closing parentheses.

6. [Fill] Complete the statement: a common use of stack in program execution is _____.

Answer: function-call management

Explanation: The call stack stores active function-call information.

7. [Fill] Complete the statement: the data structure property of a stack is _____.

Answer: linear

Explanation: A stack is a linear data structure.

8. [Fill] Complete the statement: the usual empty-stack test for a Python list named s is _____.

Answer: len(s)==0

Explanation: An empty list has length zero.

1. [T/F] True or False: the data structure property of a stack is nonlinear only.

Answer: False

Explanation: Correct answer: linear. A stack is a linear data structure.

2. [T/F] True or False: the usual empty-stack test for a Python list named s is len(s)==0.

Answer: True

Explanation: An empty list has length zero.

3. [T/F] True or False: the result of pop() on a non-empty stack is the bottom element is removed.

Answer: False

Explanation: Correct answer: the top element is returned and removed. pop removes the top item and returns it.

4. [T/F] True or False: the result of peek on a stack is the top element is observed without deletion.

Answer: True

Explanation: Peek is a non-destructive inspection operation.

5. [T/F] True or False: the main reason recursion uses a stack is to store IP addresses.

Answer: False

Explanation: Correct answer: to store active calls and return information. Each active recursive call is represented on the call stack.

6. [T/F] True or False: the postfix expression 3 5 + evaluates to is 8.

Answer: True

Explanation: Postfix 3 5 + means 3+5.

1. [VSA] Name the concept associated with: the operation that reads the top item without removing it.

Answer: peek

Explanation: Peek inspects the top element.

2. [VSA] Name the concept associated with: the condition when an item is removed from an empty stack.

Answer: underflow

Explanation: Underflow occurs when deletion is attempted on an empty stack.

3. [VSA] Name the concept associated with: the condition when a stack cannot accept another item.

Answer: overflow

Explanation: Overflow occurs when no space is available for another item in a bounded stack.

4. [VSA] Name the concept associated with: the Python list method commonly used for push.

Answer: append()

Explanation: append() adds an item to the end, which can represent the top.

5. [VSA] Name the concept associated with: the Python list method commonly used for pop.

Answer: pop()

Explanation: list.pop() removes and returns an item, commonly the top item.

6. [VSA] Name the concept associated with: the notation with operators before operands.

Answer: prefix

Explanation: Prefix places the operator before its operands.

7. [VSA] Name the concept associated with: the notation with operators between operands.

Answer: infix

Explanation: Infix is the usual human-readable arithmetic form.

1. [Short/Application] Explain the following concept in detail: the condition when a stack cannot accept another item.

Answer: overflow

Explanation: Overflow occurs when no space is available for another item in a bounded stack.

2. [Short/Application] Explain the following concept in detail: the Python list method commonly used for push.

Answer: append()

Explanation: append() adds an item to the end, which can represent the top.

3. [Short/Application] Explain the following concept: the Python list method commonly used for pop.

Answer: pop()

Explanation: list.pop() removes and returns an item, commonly the top item.

4. [Short/Application] Explain the following concept: the notation with operators before operands.

Answer: prefix

Explanation: Prefix places the operator before its operands.

5. [Short/Application] Explain the following concept: the notation with operators between operands.

Answer: infix

Explanation: Infix is the usual human-readable arithmetic form.

6. [Short/Application] Explain the following concept: the notation with operators after operands.

Answer: postfix

Explanation: Postfix is also called Reverse Polish notation.

7. [Short/Application] Explain the following concept: a common use of stack in expression processing.

Answer: parentheses matching

Explanation: Stacks can track opening parentheses and match them with closing

parentheses.

8. [Short/Application] Explain the following concept: a common use of stack in program execution.

Answer: function-call management

Explanation: The call stack stores active function-call information.

9. [Short/Application] Explain the following concept: the data structure property of a stack.

Answer: linear

LESSON 04 — QUEUE

A. Multiple Choice Questions

1. A queue follows:

A) LIFO B) FIFO C) Random D) Priority only

Answer: (B)

2. Insertion in a queue occurs at the:

A) front B) rear C) top D) middle

Answer: (B)

3. Deletion in a queue occurs at the:

A) front B) rear C) top D) middle

Answer: (A)

4. Insertion is called:

A) push B) enqueue C) pop D) dequeue

Answer: (B)

5. Deletion is called:

A) enqueue B) dequeue C) push D) peek

Answer: (B)

6. A queue with no elements is:

A) full B) empty C) overflow D) circular

Answer: (B)

7. A deque permits insertion/deletion at:

A) only front B) only rear C) both ends D) middle only

Answer: (C)

8. Which Python collection is convenient for efficient queue ends?

A) deque B) float C) tuple only D) set only

Answer: (A)

9. Deleting from an empty fixed queue causes:

A) underflow B) overflow C) collision D) recursion

Answer: (A)

10. Queue is commonly used in:

A) job scheduling B) encryption key generation only

C) syntax parsing only D) hashing only

Answer: (A)

B. Fill in the Blanks

1. Queue follows the _____ principle.

Answer: FIFO

2. Insertion into a queue is called _____.

Answer: enqueue

3. Deletion from a queue is called _____.

Answer: dequeue

4. Insertion takes place at the _____ end.

Answer: rear

5. Deletion takes place at the _____ end.

Answer: front

C. True / False

1. Queue follows FIFO.

Answer: True

2. Queue deletion normally occurs at the rear.

Answer: False

3. A deque permits operations at both ends.

Answer: True

4. BFS can use a queue.

Answer: True

5. A priority queue always removes elements strictly in arrival order.

Answer: False

D. Very Short Answer Questions

1. What is a queue?

Answer: A queue is a linear data structure in which insertion occurs at the rear and deletion occurs at the front.

2. What is FIFO?

Answer: FIFO means First In, First Out.

3. What is enqueue?

Answer: Enqueue adds an element to the rear of a queue.

4. What is dequeue?

Answer: Dequeue removes an element from the front of a queue.

5. What is a deque?

Answer: A deque is a double-ended queue in which insertion and deletion can be performed at either end.

E. Short Answer Questions

1. Explain enqueue and dequeue.

Answer: Enqueue inserts an item at the rear. Dequeue removes an item from the front, preserving FIFO order.

2. Differentiate queue and stack.

Answer: A queue inserts at the rear and deletes at the front using FIFO. A stack inserts and deletes at the top using LIFO.

3. Explain front and rear.

Answer: Front identifies the element position from which deletion occurs. Rear identifies the position where new elements are inserted.

4. Explain applications of queue.

Answer: Queues are used in CPU/job scheduling, printer queues, buffering, breadth-first search and service systems.

5. Differentiate queue and deque.

Answer: A normal queue restricts insertion to rear and deletion to front. A deque permits insertion and deletion at both ends.

F. Long Answer / Programming Questions

1. Write a Python program to implement a queue.

Answer: Maintain a list/deque. Enqueue appends at the rear; dequeue removes from the front. Include checks for an empty queue and operations such as peek/display.

2. Explain circular queue.

Answer: A circular queue treats the last position as connected to the first. Front and rear wrap around using modular arithmetic, allowing freed positions to be reused efficiently.

3. Explain priority queue.

Answer: A priority queue associates each item with a priority. Removal is based on priority rather than simple arrival order; items of equal priority can follow FIFO depending on implementation.

4. Explain deque with operations.

Answer: A deque supports insertFront, insertRear, deletionFront and deletionRear. It can behave like a queue or stack depending on which ends are used.

5. Explain queue in BFS.

Answer: BFS explores vertices level by level. Newly discovered vertices are enqueued, and the next vertex to process is dequeued from the front, giving breadth-first order.

Additional Questions

1. [MCQ] Which of the following is associated with the access principle of a queue?

A) FIFO B) LIFO C) random access D) priority only

Answer: FIFO

Explanation: FIFO means First In, First Out.

2. [MCQ] Which of the following is associated with the operation that inserts an item into a queue?

A) enqueue B) dequeue C) push D) pop

Answer: enqueue

Explanation: Enqueue adds an item, normally at the rear.

3. [MCQ] Which of the following is associated with the operation that removes an item from a queue?

A) dequeue B) enqueue C) peek D) push

Answer: dequeue

Explanation: Dequeue removes the front item.

4. [MCQ] Which of the following is associated with the end normally used for insertion?

A) rear B) front C) middle D) top

Answer: rear

Explanation: New queue elements are added at the rear.

5. [MCQ] Which of the following is associated with the end normally used for deletion?

A) front B) rear C) middle D) top

Answer: front

Explanation: The oldest queue element is removed from the front.

6. [MCQ] Which of the following is associated with the condition when deletion is attempted on an empty queue?

A) underflow B) overflow C) collision D) syntax error

Answer: underflow

Explanation: Queue underflow occurs when no item is available to remove.

7. [MCQ] Which of the following is associated with the condition when a bounded queue is full and insertion is attempted?

A) overflow B) underflow C) deadlock D) collision

Answer: overflow

Explanation: Queue overflow indicates that the available capacity is exhausted.

8. [MCQ] Which of the following is associated with a deque that permits insertion and deletion at both ends?

A) double-ended queue B) priority queue C) circular queue

D) simple stack

Answer: double-ended queue

Explanation: Deque is short for double-ended queue.

9. [MCQ] Which of the following is associated with a queue in which service is based on priority?

A) priority queue B) circular queue C) simple queue D) stack

Answer: priority queue

Explanation: Priority queues remove elements according to priority rather than only

arrival order.

10. [MCQ] Which of the following is associated with a queue whose last position is logically connected to the first?

A) circular queue B) linear queue C) priority queue D) stack

Answer: circular queue

Explanation: Circular queues reuse positions by wrapping around.

11. [MCQ] Which of the following is associated with a Python structure often used to implement a simple queue?

A) list B) tuple C) set D) string only

Answer: list

Explanation: A list can model queue operations, although deque is efficient for front deletion.

12. [MCQ] Which of the following is associated with the Python module providing an efficient deque?

A) collections B) queueing C) stacklib D) network

Answer: collections

Explanation: collections. Deque supports efficient operations at both ends.

13. [MCQ] Which of the following is associated with the usual initial value of front and rear in a conceptual empty queue?

A) -1 B) 0 only C) 1 D) None always

Answer: -1

Explanation: Array-based teaching implementations commonly initialize front and rear to -1.

14. [MCQ] Which of the following is associated with the main application of a queue in scheduling?

A) first-come service order B) reverse order

C) random service D) last-in service

Answer: first-come service order

Explanation: Queues model tasks waiting for service.

15. [MCQ] Which of the following is associated with the traversal that commonly uses a queue?

A) breadth-first search B) depth-first search

C) binary search D) selection sort

Answer: breadth-first search

Explanation: BFS explores nodes level by level using a queue.

1. [Fill] Complete the statement: a deque that permits insertion and deletion at both ends is _____.

Answer: double-ended queue

Explanation: Deque is short for double-ended queue.

2. [Fill] Complete the statement: a queue in which service is based on priority is _____.

Answer: priority queue

Explanation: Priority queues remove elements according to priority rather than only arrival order.

3. [Fill] Complete the statement: a queue whose last position is logically connected to the first is _____.

Answer: circular queue

Explanation: Circular queues reuse positions by wrapping around.

4. [Fill] Complete the statement: a Python structure often used to implement a simple queue is _____.

Answer: list

Explanation: A list can model queue operations, although deque is efficient for front deletion.

5. [Fill] Complete the statement: the Python module providing an efficient deque is _____.

Answer: collections

Explanation: collections. Deque supports efficient operations at both ends.

6. [Fill] Complete the statement: the usual initial value of front and rear in a conceptual empty queue is _____.

Answer: -1

Explanation: Array-based teaching implementations commonly initialize front and rear to -1.

7. [Fill] Complete the statement: the main application of a queue in scheduling is _____.

Answer: first-come service order

Explanation: Queues model tasks waiting for service.

8. [Fill] Complete the statement: the traversal that commonly uses a queue is _____.

Answer: breadth-first search

Explanation: BFS explores nodes level by level using a queue.

1. [T/F] True or False: the main application of a queue in scheduling is reverse order.

Answer: False

Explanation: Correct answer: first-come service order. Queues model tasks waiting for service.

2. [T/F] True or False: the traversal that commonly uses a queue is breadth-first search.

Answer: True

Explanation: BFS explores nodes level by level using a queue.

3. [T/F] True or False: the operation that inspects the front without deleting it is pop.

Answer: False

Explanation: Correct answer: peek/front access. A peek-like operation can inspect the

front item.

4. [T/F] True or False: the basic number of core operations in a simple queue is two: enqueue and dequeue.

Answer: True

Explanation: Insertion and deletion are the two fundamental queue operations.

5. [T/F] True or False: a real-life example of FIFO behavior is undo history.

Answer: False

Explanation: Correct answer: people waiting in a line. The person who arrives first is served first in a normal line.

6. [T/F] True or False: the queue principle contrasted with stack principle is FIFO rather than LIFO.

Answer: True

Explanation: Queues normally follow FIFO while stacks follow LIFO.

1. [VSA] Name the concept associated with: the end normally used for insertion.

Answer: rear

Explanation: New queue elements are added at the rear.

2. [VSA] Name the concept associated with: the end normally used for deletion.

Answer: front

Explanation: The oldest queue element is removed from the front.

3. [VSA] Name the concept associated with: the condition when deletion is attempted on an empty queue.

Answer: underflow

Explanation: Queue underflow occurs when no item is available to remove.

4. [VSA] Name the concept associated with: the condition when a bounded queue is full and insertion is attempted.

Answer: overflow

Explanation: Queue overflow indicates that the available capacity is exhausted.

5. [VSA] Name the concept associated with: a deque that permits insertion and deletion at both ends.

Answer: double-ended queue

Explanation: Deque is short for double-ended queue.

6. [VSA] Name the concept associated with: a queue in which service is based on priority.

Answer: priority queue

Explanation: Priority queues remove elements according to priority rather than only arrival order.

7. [VSA] Name the concept associated with: a queue whose last position is logically connected to the first.

Answer: circular queue

Explanation: Circular queues reuse positions by wrapping around.

1. [Short/Application] Explain the following concept in detail: the condition when deletion is attempted on an empty queue.

Answer: underflow

Explanation: Queue underflow occurs when no item is available to remove.

2. [Short/Application] Explain the following concept in detail: the condition when a bounded queue is full and insertion is attempted.

Answer: overflow

Explanation: Queue overflow indicates that the available capacity is exhausted.

3. [Short/Application] Explain the following concept in detail: a deque that permits insertion and deletion at both ends.

Answer: double-ended queue

Explanation: Deque is short for double-ended queue.

4. [Short/Application] Explain the following concept in detail: a queue in which service is based on priority.

Answer: priority queue

Explanation: Priority queues remove elements according to priority rather than only arrival order.

5. [Short/Application] Explain the following concept in detail: a queue whose last position is logically connected to the first.

Answer: circular queue

Explanation: Circular queues reuse positions by wrapping around.

6. [Short/Application] Explain the following concept in detail: a Python structure often used to implement a simple queue.

Answer: list

Explanation: A list can model queue operations, although deque is efficient for front deletion.

7. [Short/Application] Explain the following concept in detail: the Python module providing an efficient deque.

Answer: collections

Explanation: collections. deque supports efficient operations at both ends.

8. [Short/Application] Explain the following concept in detail: the usual initial value of front and rear in a conceptual empty queue.

Answer: -1

Explanation: Array-based teaching implementations commonly initialize front and rear to -1.

9. [Short/Application] Explain the following concept in detail: the main application of a queue in scheduling.

Answer: first-come service order

Explanation: Queues model tasks waiting for service.

LESSON 05 — SORTING

A. Multiple Choice Questions

1. Sorting means:

A) Arranging elements in an order B) Deleting elements

C) Encrypting data D) Searching only

Answer: (A)

2. Bubble sort repeatedly compares:

A) Adjacent elements B) First and last only

C) Random elements D) Hash values

Answer: (A)

3. Selection sort selects the:

A) Smallest element from unsorted part for ascending order

B) Largest always C) Middle element D) Random element

Answer: (A)

4. Insertion sort builds a:

A) Sorted portion incrementally

B) Hash table C) Queue only D) Tree only

Answer: (A)

5. Basic bubble, selection and insertion sorts in the prescribed chapter are:

A) O(1) B) O(n) C) $O(n^2)$ D) O(log n)

Answer: (C)

6. A stable sort preserves the relative order of:

A) Equal-key elements B) All elements

C) Only minimum elements D) Only maximum elements

Answer: (A)

7. Insertion sort is particularly useful for:

A) Nearly sorted/small lists B) Encrypted data only

C) Networks only D) Databases only

Answer: (A)

8. Selection sort generally performs:

A) Relatively few swaps B) No comparisons

C) Hashing D) Recursive calls only

Answer: (A)

9. Bubble sort works through multiple:

A) Passes B) Files C) Queues D) Networks

Answer: (A)

10. Time complexity describes growth of:

A) Algorithmic work with input size

B) Screen size C) File extension D) IP address

Answer: (A)

B. Fill in the Blanks

1. The process of arranging data in order is called _____.

Answer: sorting

2. Bubble sort compares _____ elements.

Answer: adjacent

3. Selection sort selects the _____ element from the unsorted portion for ascending order.

Answer: smallest

4. Insertion sort inserts an item into the correct position in the _____ portion.

Answer: sorted

5. The basic sorting algorithms discussed have _____ worst-case time complexity.

Answer: $O(n^2)$

C. True / False

1. Bubble sort repeatedly compares adjacent elements.

Answer: True

2. Selection sort repeatedly selects an element from the unsorted part.

Answer: True

3. Insertion sort cannot work on nearly sorted data efficiently.

Answer: False

4. A stable sort preserves the relative order of equal-key records.

Answer: True

5. The prescribed basic sorting algorithms have constant time complexity.

Answer: False

D. Very Short Answer Questions

1. Define sorting.

Answer: Sorting is the process of placing or rearranging a collection of elements into a particular order.

2. What is a pass?

Answer: A pass is one complete traversal or major iteration through the relevant portion of a list during a sorting algorithm.

3. What is stability?

Answer: Stability means equal-key elements retain their relative order after sorting.

4. Name three sorting methods in the chapter.

Answer: Bubble sort, selection sort and insertion sort.

5. What is time complexity?

Answer: It describes how the computational work of an algorithm grows with input size.

E. Short Answer Questions

1. Explain bubble sort.

Answer: Compare adjacent elements and swap them when they are in the wrong order. Repeated passes move extreme elements toward their final positions.

2. Explain selection sort.

Answer: At each pass, find the smallest element in the unsorted portion and swap it with the first element of that portion.

3. Explain insertion sort.

Answer: Maintain a sorted prefix and insert each next element into its correct position by shifting larger elements to the right.

4. Why is insertion sort adaptive?

Answer: If the list is already or nearly sorted, only a small number of shifts/comparisons are needed, so it can perform much better than on a reverse/random list.

5. Differentiate bubble and selection sort.

Answer: Bubble sort repeatedly swaps adjacent out-of-order elements. Selection sort finds a minimum in the unsorted region and performs a swap with its boundary element.

F. Long Answer / Programming Questions

1. Trace bubble sort on [5, 1, 4, 2].

Answer: Pass 1: [1,4,2,5]; Pass 2: [1,2,4,5]; Pass 3 makes no change. The final sorted list is [1,2,4,5].

2. Explain selection sort with an example.

Answer: For [4,2,5,1], select 1 and swap with 4 → [1,2,5,4]; next select 2 → unchanged; select 4 and swap with 5 → [1,2,4,5].

3. Explain insertion sort with an example.

Answer: For [5,2,4,1], insert 2 before 5 → [2,5,4,1]; insert 4 → [2,4,5,1]; insert 1 → [1,2,4,5].

4. Compare bubble, selection and insertion sort.

Answer: All are comparison-based and have quadratic worst-case complexity in the prescribed treatment. Bubble uses adjacent swaps, selection chooses minima, and insertion grows a sorted prefix. Their practical behaviour and number of swaps differ.

5. Explain complexity in the prescribed chapter.

Answer: A loop over n items is linear; a nested loop is quadratic. The basic bubble, selection and insertion implementations contain nested iteration, so their time complexity is treated as $O(n^2)$.

Additional Questions

1. [MCQ] Which of the following is associated with the sorting algorithm that swaps adjacent out-of-order elements?

A) Bubble Sort

B) Selection Sort

C) Insertion Sort

D) Binary Search

Answer: Bubble Sort

Explanation: Bubble Sort repeatedly compares adjacent elements.

2. [MCQ] Which of the following is associated with the sorting algorithm that selects the minimum from the unsorted portion?

A) Selection Sort

B) Bubble Sort

C) Insertion Sort

D) Linear Search

Answer: Selection Sort

Explanation: Selection Sort selects an extreme element and places it in position.

3. [MCQ] Which of the following is associated with the sorting algorithm that inserts each item into an already sorted prefix?

A) Insertion Sort

B) Selection Sort

C) Bubble Sort

D) Hashing

Answer: Insertion Sort

Explanation: Insertion Sort grows a sorted portion one element at a time.

4. [MCQ] Which of the following is associated with the usual worst-case time complexity of Bubble Sort?

A) $O(n^2)$

B) O(1)

C) O(log n)

D) O(n log n)

Answer: $O(n^2)$

Explanation: The nested comparisons in the worst case give quadratic time.

5. [MCQ] Which of the following is associated with the usual worst-case time complexity of Selection Sort?

A) $O(n^2)$

B) O(n)

C) O(log n)

D) O(1)

Answer: $O(n^2)$

Explanation: Selection Sort scans the remaining unsorted portion for each position.

6. [MCQ] Which of the following is associated with the usual worst-case time complexity of Insertion Sort?

A) $O(n^2)$

B) O(1)

C) O(log n)

D) O(n)

Answer: $O(n^2)$

Explanation: In the reverse-sorted case, many shifts are required.

7. [MCQ] Which of the following is associated with the property of preserving relative order of equal keys?

A) stable

B) adaptive

C) recursive

D) in-place only

Answer: stable

Explanation: A stable sort keeps equal-key records in their original relative order.

8. [MCQ] Which of the following is associated with the sorting algorithm among the basic three that is particularly effective on nearly sorted data?

A) Insertion Sort

B) Selection Sort

C) Binary Search

D) Hashing

Answer: Insertion Sort

Explanation: Insertion Sort can perform close to linear time when few shifts are needed.

9. [MCQ] Which of the following is associated with the main operation in Selection Sort?

A) select an extreme element from the unsorted portion

B) compare adjacent only

C) divide into hashes

D) search by midpoint

Answer: select an extreme element from the unsorted portion

Explanation: Selection Sort repeatedly selects the minimum or maximum.

10. [MCQ] Which of the following is associated with the main operation in Bubble Sort?

A) compare adjacent elements and swap when needed

B) select a global minimum only

C) hash each key

D) split around a pivot

Answer: compare adjacent elements and swap when needed

Explanation: Adjacent comparisons characterize Bubble Sort.

11. [MCQ] Which of the following is associated with the sorted portion in Insertion Sort?

A) the prefix already processed

B) the suffix only

C) random positions

D) the hash table

Answer: the prefix already processed

Explanation: Insertion Sort maintains a sorted prefix.

12. [MCQ] Which of the following is associated with the measure used to describe growth of sorting time?

A) time complexity

B) file size only

C) IP address

D) bandwidth

Answer: time complexity

Explanation: Time complexity describes how running time grows with input size.

13. [MCQ] Which of the following is associated with a sort that orders values from smallest to largest?

A) ascending order

B) descending order

C) random order

D) cyclic order

Answer: ascending order

Explanation: Ascending means increasing numeric or lexicographic order.

14. [MCQ] Which of the following is associated with a sort that orders values from largest to smallest?

A) descending order

B) ascending order

C) random order

D) FIFO order

Answer: descending order

Explanation: Descending means decreasing order.

15. [MCQ] Which of the following is associated with the number of major passes in basic Selection Sort?

A) approximately n-1

B) n^2 always

C) log n

D) 1 only

Answer: approximately n-1

Explanation: Selection Sort places one element per pass, typically n-1 passes.

16. [MCQ] Which of the following is associated with the condition for an already sorted list to be the best case for many adaptive sorts?

A) few or no movements are required

B) maximum inversions

C) all values duplicated

D) no comparisons possible

Answer: few or no movements are required

Explanation: Adaptive algorithms exploit existing order.

17. [MCQ] Which of the following is associated with a comparison-based sorting algorithm compares?

A) keys/elements to determine order

B) IP addresses only

C) files only

D) ports only

Answer: keys/elements to determine order

Explanation: Comparison sorts establish order through comparisons.

1. [Fill] Complete the statement: the sorting algorithm among the basic three that is particularly effective on nearly sorted data is _____.

Answer: Insertion Sort

Explanation: Insertion Sort can perform close to linear time when few shifts are needed.

2. [Fill] Complete the statement: the main operation in Selection Sort is _____.

Answer: select an extreme element from the unsorted portion

Explanation: Selection Sort repeatedly selects the minimum or maximum.

3. [Fill] Complete the statement: the main operation in Bubble Sort is _____.

Answer: compare adjacent elements and swap when needed

Explanation: Adjacent comparisons characterize Bubble Sort.

4. [Fill] Complete the statement: the sorted portion in Insertion Sort is _____.

Answer: the prefix already processed

Explanation: Insertion Sort maintains a sorted prefix.

5. [Fill] Complete the statement: the measure used to describe growth of sorting time is _____.

Answer: time complexity

Explanation: Time complexity describes how running time grows with input size.

6. [Fill] Complete the statement: a sort that orders values from smallest to largest is _____.

Answer: ascending order

Explanation: Ascending means increasing numeric or lexicographic order.

7. [Fill] Complete the statement: a sort that orders values from largest to smallest is _____.

Answer: descending order

Explanation: Descending means decreasing order.

8. [Fill] Complete the statement: the number of major passes in basic Selection Sort is _____.

Answer: approximately n-1

Explanation: Selection Sort places one element per pass, typically n-1 passes.

9. [Fill] Complete the statement: the condition for an already sorted list to be the best case for many adaptive sorts is _____.

Answer: few or no movements are required

Explanation: Adaptive algorithms exploit existing order.

1. [T/F] True or False: a sort that orders values from largest to smallest is ascending order.

Answer: False

Explanation: Correct answer: descending order. Descending means decreasing order.

2. [T/F] True or False: the number of major passes in basic Selection Sort is approximately n-1.

Answer: True

Explanation: Selection Sort places one element per pass, typically n-1 passes.

3. [T/F] True or False: the condition for an already sorted list to be the best case for many adaptive sorts is maximum inversions.

Answer: False

Explanation: Correct answer: few or no movements are required. Adaptive algorithms exploit existing order.

4. [T/F] True or False: a comparison-based sorting algorithm compares is keys/elements to determine order.

Answer: True

Explanation: Comparison sorts establish order through comparisons.

5. [T/F] True or False: the common purpose of sorting is encrypt data.

Answer: False

Explanation: Correct answer: arrange data according to an ordering criterion. Sorting organizes records for easier processing and searching.

6. [T/F] True or False: why sorting can help binary search is binary search requires ordered data.

Answer: True

Explanation: Binary search relies on sorted order.

7. [T/F] True or False: the result of sorting [5,2,4] ascending is [5,4,2].

Answer: False

Explanation: Correct answer: [2, 4, 5]. Ascending order places values from smallest to largest.

1. [VSA] Name the concept associated with: the usual worst-case time complexity of Bubble Sort.

Answer: $O(n^2)$

Explanation: The nested comparisons in the worst case give quadratic time.

2. [VSA] Name the concept associated with: the usual worst-case time complexity of Selection Sort.

Answer: $O(n^2)$

Explanation: Selection Sort scans the remaining unsorted portion for each position.

3. [VSA] Name the concept associated with: the usual worst-case time complexity of Insertion Sort.

Answer: $O(n^2)$

Explanation: In the reverse-sorted case, many shifts are required.

4. [VSA] Name the concept associated with: the property of preserving relative order of equal keys.

Answer: stable

Explanation: A stable sort keeps equal-key records in their original relative order.

5. [VSA] Name the concept associated with: the sorting algorithm among the basic three that is particularly effective on nearly sorted data.

Answer: Insertion Sort

Explanation: Insertion Sort can perform close to linear time when few shifts are needed.

6. [VSA] Name the concept associated with: the main operation in Selection Sort.

Answer: select an extreme element from the unsorted portion

Explanation: Selection Sort repeatedly selects the minimum or maximum.

7. [VSA] Name the concept associated with: the main operation in Bubble Sort.

Answer: compare adjacent elements and swap when needed

Explanation: Adjacent comparisons characterize Bubble Sort.

8. [VSA] Name the concept associated with: the sorted portion in Insertion Sort.

Answer: the prefix already processed

Explanation: Insertion Sort maintains a sorted prefix.

1. [Short/Application] Explain the following concept in detail: the usual worst-case time complexity of Insertion Sort.

Answer: $O(n^2)$

Explanation: In the reverse-sorted case, many shifts are required.

2. [Short/Application] Explain the following concept in detail: the property of preserving relative order of equal keys.

Answer: stable

Explanation: A stable sort keeps equal-key records in their original relative order.

3. [Short/Application] Explain the following concept in detail: the sorting algorithm among the basic three that is particularly effective on nearly sorted data.

Answer: Insertion Sort

Explanation: Insertion Sort can perform close to linear time when few shifts are needed.

4. [Short/Application] Explain the following concept in detail: the main operation in Selection Sort.

Answer: select an extreme element from the unsorted portion

Explanation: Selection Sort repeatedly selects the minimum or maximum.

5. [Short/Application] Explain the following concept in detail: the main operation in Bubble Sort.

Answer: compare adjacent elements and swap when needed

Explanation: Adjacent comparisons characterize Bubble Sort.

6. [Short/Application] Explain the following concept in detail: the sorted portion in Insertion Sort.

Answer: the prefix already processed

Explanation: Insertion Sort maintains a sorted prefix.

7. [Short/Application] Explain the following concept in detail: the measure used to describe growth of sorting time.

Answer: time complexity

Explanation: Time complexity describes how running time grows with input size.

8. [Short/Application] Explain the following concept in detail: a sort that orders values from smallest to largest.

Answer: ascending order

Explanation: Ascending means increasing numeric or lexicographic order.

9. [Short/Application] Explain the following concept in detail: a sort that orders values from largest to smallest.

Answer: descending order

Explanation: Descending means decreasing order.

LESSON 06 — SEARCHING

A. Multiple Choice Questions

1. Searching means:

A) Locating an element in a collection

B) Sorting only

C) Encrypting

D) Deleting

Answer: (A)

2. Linear search checks elements:

A) One by one

B) By halving only

C) By hashing only

D) Randomly

Answer: (A)

3. Binary search requires the list to be:

A) Sorted

B) Empty

C) Encrypted

D) Circular

Answer: (A)

4. Best-case linear search is:

A) O(1)

B) O(n)

C) O(log n)

D) $O(n^2)$

Answer: (A)

5. Worst-case linear search is:

A) O(1)

B) O(n)

C) O(log n)

D) $O(n^2)$

Answer: (B)

6. Worst-case binary search is:

A) O(n)

B) O(log n)

C) $O(n^2)$

D) O(1)

Answer: (B)

7. Hashing uses a:

A) Hash function

B) Sort function

C) Queue only

D) Loop only

Answer: (A)

8. Two keys mapping to the same hash slot cause a:

A) Collision

B) Stack

C) Pass

D) Token

Answer: (A)

9. A simple remainder hash is:

A) key % table_size

B) key + table_size

C) key * table_size

D) key / 0

Answer: (A)

10. Which is a collision-resolution approach?

A) Chaining

B) Formatting

C) Sorting only

D) Printing

Answer: (A)

B. Fill in the Blanks

1. Linear search examines elements _____.

Answer: sequentially/one by one

2. Binary search works on a _____ list.

Answer: sorted

3. Worst-case linear search is _____.

Answer: O(n)

4. Worst-case binary search is _____.

Answer: O(log n)

5. When two keys map to the same slot, a _____ occurs.

Answer: collision

C. True / False

1. Binary search can be applied directly to an arbitrary unsorted list.

Answer: False

2. Linear search can work on an unsorted list.

Answer: True

3. Hashing can produce collisions.

Answer: True

4. Binary search repeatedly reduces the search interval.

Answer: True

5. A hash table position can be calculated using a hash function.

Answer: True

D. Very Short Answer Questions

1. What is linear search?

Answer: It checks elements sequentially until the target is found or the collection ends.

2. What is binary search?

Answer: It searches a sorted list by repeatedly comparing with the middle element and discarding half of the remaining interval.

3. What is a hash function?

Answer: A formula that maps a key to an index in a hash table.

4. What is collision?

Answer: A collision occurs when two or more keys map to the same hash-table position.

5. Name two collision-resolution techniques.

Answer: Chaining and open addressing are common techniques.

E. Short Answer Questions

1. Compare linear and binary search.

Answer: Linear search examines items sequentially and needs no sorting. Binary search requires sorted data and repeatedly halves the search range, giving O(log n) worst-case comparisons.

2. Explain binary search.

Answer: Set low and high bounds, find the middle, compare the target with the middle value, and continue in the appropriate half until found or the range becomes empty.

3. Explain hashing.

Answer: Hashing calculates a table index from a key using a hash function, allowing direct or near-direct access when the distribution and collision handling are effective.

4. Explain collision resolution.

Answer: When multiple keys map to one slot, methods such as chaining store multiple entries through a linked/bucket structure, while open addressing searches other slots according to a probing rule.

5. State the time complexities.

Answer: Linear search: best O(1), worst O(n). Binary search: best O(1), worst O(log n), assuming the required sorted structure is available.

F. Long Answer / Programming Questions

1. Write a Python program for linear search.

Answer: Traverse the list with an index; if list[i] equals the target, report the position and stop. If the loop ends, report that the element is absent.

2. Write a Python program for binary search.

Answer: Maintain low=0 and high=len(list)-1; repeatedly compute mid, compare target with list[mid], and update low/high until found or low exceeds high.

3. Trace binary search on [10,20,30,40,50,60,70] for 60.

Answer: Middle is 40; 60 is larger, so search the upper half. Middle there is 60, so the element is found.

4. Explain hashing by remainder method.

Answer: For table size m, calculate h(key)=key % m. Place the key at that index when free; if occupied, apply a collision-resolution strategy.

5. Compare searching techniques.

Answer: Linear search is simple and works on unsorted data but may inspect every item. Binary search is faster for sorted data because it halves the interval. Hashing can provide very fast direct lookup when a suitable hash table and collision strategy are available.

Additional Questions

1. [MCQ] Which of the following is associated with the search method that checks items sequentially?

A) Linear Search

B) Binary Search

C) Hashing

D) Selection Sort

Answer: Linear Search

Explanation: Linear Search examines elements one by one.

2. [MCQ] Which of the following is associated with the search method that repeatedly halves a sorted search interval?

A) Binary Search

B) Linear Search

C) Bubble Sort

D) Hashing

Answer: Binary Search

Explanation: Binary Search uses the midpoint to discard half the search range.

3. [MCQ] Which of the following is associated with the prerequisite for Binary Search in the prescribed treatment?

A) a sorted list

B) an empty list only

C) a hash table only

D) a queue only

Answer: a sorted list

Explanation: Binary Search depends on sorted ordering.

4. [MCQ] Which of the following is associated with the best-case time complexity of Linear Search?

A) O(1)

B) O(log n)

C) $O(n^2)$

D) O(n log n)

Answer: O(1)

Explanation: The target may be found at the first position.

5. [MCQ] Which of the following is associated with the worst-case time complexity of Linear Search?

A) O(n)

B) O(1)

C) O(log n)

D) $O(n^2)$

Answer: O(n)

Explanation: The target may be at the end or absent.

6. [MCQ] Which of the following is associated with the worst-case time complexity of Binary Search?

A) O(log n)

B) O(n)

C) $O(n^2)$

D) O(1)

Answer: O(log n)

Explanation: The search interval is halved each iteration.

7. [MCQ] Which of the following is associated with the function that maps a key to a table position?

A) hash function

B) sort function

C) search loop

D) index printer

Answer: hash function

Explanation: A hash function maps a key to a hash-table slot.

8. [MCQ] Which of the following is associated with the condition when two keys map to the same hash slot?

A) collision

B) overflow only

C) underflow

D) sorting

Answer: collision

Explanation: A collision occurs when multiple keys receive the same slot.

9. [MCQ] Which of the following is associated with a collision-resolution method that stores a list at each slot?

A) chaining

B) selection

C) binary search

D) FIFO

Answer: chaining

Explanation: Chaining maintains a collection of entries for a slot.

10. [MCQ] Which of the following is associated with a collision-resolution approach that searches for another free slot?

A) open addressing

B) file handling

C) sorting

D) recursion

Answer: open addressing

Explanation: Open addressing probes other positions in the table.

11. [MCQ] Which of the following is associated with the remainder method hash formula for key k and table size m?

A) k mod m

B) k+m

C) k/m without remainder

D) m-k

Answer: k mod m

Explanation: The remainder method uses k modulo m.

12. [MCQ] Which of the following is associated with the purpose of a hash table?

A) fast average-case key lookup

B) maintain FIFO order

C) sort every list

D) store only images

Answer: fast average-case key lookup

Explanation: Hash tables support efficient average-case lookup by key.

13. [MCQ] Which of the following is associated with the search interval after finding target greater than midpoint in ascending Binary Search?

A) the right half

B) the left half

C) the whole list

D) an unrelated list

Answer: the right half

Explanation: If target > midpoint value, the right half is retained.

14. [MCQ] Which of the following is associated with the search interval after finding target smaller than midpoint in ascending Binary Search?

A) the left half

B) the right half

C) the whole list

D) the hash table

Answer: the left half

Explanation: If target < midpoint value, the left half is retained.

15. [MCQ] Which of the following is associated with the condition when Binary Search fails to find a value?

A) the search interval becomes empty

B) the list is automatically sorted

C) the target is inserted

D) the table doubles

Answer: the search interval becomes empty

Explanation: An unsuccessful search terminates when no positions remain.

1. [Fill] Complete the statement: the condition when two keys map to the same hash slot is _____.

Answer: collision

Explanation: A collision occurs when multiple keys receive the same slot.

2. [Fill] Complete the statement: a collision-resolution method that stores a list at each slot is _____.

Answer: chaining

Explanation: Chaining maintains a collection of entries for a slot.

3. [Fill] Complete the statement: a collision-resolution approach that searches for another free slot is _____.

Answer: open addressing

Explanation: Open addressing probes other positions in the table.

4. [Fill] Complete the statement: the remainder method hash formula for key k and table size m is _____.

Answer: k mod m

Explanation: The remainder method uses k modulo m.

5. [Fill] Complete the statement: the purpose of a hash table is _____.

Answer: fast average-case key lookup

Explanation: Hash tables support efficient average-case lookup by key.

6. [Fill] Complete the statement: the search interval after finding target greater than midpoint in ascending Binary Search is _____.

Answer: the right half

Explanation: If target > midpoint value, the right half is retained.

7. [Fill] Complete the statement: the search interval after finding target smaller than midpoint in ascending Binary Search is _____.

Answer: the left half

Explanation: If target < midpoint value, the left half is retained.

8. [Fill] Complete the statement: the condition when Binary Search fails to find a value is _____.

Answer: the search interval becomes empty

Explanation: An unsuccessful search terminates when no positions remain.

1. [T/F] True or False: the search interval after finding target smaller than midpoint in ascending Binary Search is the right half.

Answer: False

Explanation: Correct answer: the left half. If target < midpoint value, the left half is retained.

2. [T/F] True or False: the condition when Binary Search fails to find a value is the search interval becomes empty.

Answer: True

Explanation: An unsuccessful search terminates when no positions remain.

3. [T/F] True or False: the common indexing benefit of hashing is guaranteed sorting.

Answer: False

Explanation: Correct answer: direct access by computed slot. Hashing computes a location from the key.

4. [T/F] True or False: the main difference between Linear and Binary Search is Linear checks sequentially; Binary halves a sorted range.

Answer: True

Explanation: Their search strategies and prerequisites differ.

5. [T/F] True or False: the value of log2(8) is 2.

Answer: False

Explanation: Correct answer: 3. $2^3 = 8$.

6. [T/F] True or False: the result of searching for 7 in [2,4,7,9] using Binary Search is found.

Answer: True

Explanation: 7 exists in the sorted list.

1. [VSA] Name the concept associated with: the best-case time complexity of Linear Search.

Answer: O(1)

Explanation: The target may be found at the first position.

2. [VSA] Name the concept associated with: the worst-case time complexity of Linear Search.

Answer: O(n)

Explanation: The target may be at the end or absent.

3. [VSA] Name the concept associated with: the worst-case time complexity of Binary Search.

Answer: O(log n)

Explanation: The search interval is halved each iteration.

4. [VSA] Name the concept associated with: the function that maps a key to a table position.

Answer: hash function

Explanation: A hash function maps a key to a hash-table slot.

5. [VSA] Name the concept associated with: the condition when two keys map to the same hash slot.

Answer: collision

Explanation: A collision occurs when multiple keys receive the same slot.

6. [VSA] Name the concept associated with: a collision-resolution method that stores a list at each slot.

Answer: chaining

Explanation: Chaining maintains a collection of entries for a slot.

7. [VSA] Name the concept associated with: a collision-resolution approach that searches for another free slot.

Answer: open addressing

Explanation: Open addressing probes other positions in the table.

1. [Short/Application] Explain the following concept in detail: the worst-case time complexity of Binary Search.

Answer: O(log n)

Explanation: The search interval is halved each iteration.

2. [Short/Application] Explain the following concept in detail: the function that maps a key to a table position.

Answer: hash function

Explanation: A hash function maps a key to a hash-table slot.

3. [Short/Application] Explain the following concept in detail: the condition when two

keys map to the same hash slot.

Answer: collision

Explanation: A collision occurs when multiple keys receive the same slot.

4. [Short/Application] Explain the following concept in detail: a collision-resolution method that stores a list at each slot.

Answer: chaining

Explanation: Chaining maintains a collection of entries for a slot.

5. [Short/Application] Explain the following concept in detail: a collision-resolution approach that searches for another free slot.

Answer: open addressing

Explanation: Open addressing probes other positions in the table.

6. [Short/Application] Explain the following concept in detail: the remainder method hash formula for key k and table size m.

Answer: k mod m

Explanation: The remainder method uses k modulo m.

7. [Short/Application] Explain the following concept in detail: the purpose of a hash table.

Answer: fast average-case key lookup

Explanation: Hash tables support efficient average-case lookup by key.

8. [Short/Application] Explain the following concept in detail: the search interval after finding target greater than midpoint in ascending Binary Search.

Answer: the right half

Explanation: If target > midpoint value, the right half is retained.

9. [Short/Application] Explain the following concept in detail: the search interval after finding target smaller than midpoint in ascending Binary Search.

Answer: the left half

Explanation: If target < midpoint value, the left half is retained.

LESSON 07 — UNDERSTANDING DATA

A. **Multiple Choice Questions**

1. Data arranged in rows and columns with a defined schema is:

A) Structured

B) Unstructured

C) Random

D) Encrypted

Answer: (A)

2. Images, audio and video are commonly examples of:

A) Unstructured data

B) Only structured data

C) Primary keys

D) SQL commands

Answer: (A)

3. JSON/XML commonly represent:

A) Semi-structured data

B) Only binary data

C) Hardware

D) Malware

Answer: (A)

4. The first stage in a typical data-processing cycle is:

A) Input/capture

B) Deletion

C) Sorting only

D) Encryption

Answer: (A)

5. The arithmetic average is called:

A) Mean

B) Mode

C) Range

D) Median only

Answer: (A)

6. The middle value after ordering is:

A) Median

B) Mean

C) Mode

D) Range

Answer: (A)

7. The most frequently occurring value is:

A) Mode

B) Mean

C) Median

D) Range

Answer: (A)

8. Range equals:

A) Maximum - Minimum

B) Mean - Mode

C) Sum/Count

D) Maximum + Minimum

Answer: (A)

9. Standard deviation measures:

A) Dispersion around the mean

B) Only maximum value

C) File size

D) Network speed

Answer: (A)

10. Retrieving stored data means:

A) Data retrieval

B) Data deletion

C) Data encryption

D) Data sorting only

Answer: (A)

B. **Fill in the Blanks**

1. Data arranged according to a predefined model is called _____ data.

Answer: structured

2. JSON is commonly considered _____-structured data.

Answer: semi

3. The average of values is called the _____.

Answer: mean

4. The most frequent value is the _____.

Answer: mode

5. Range is maximum value minus the _____ value.

Answer: minimum

C. True / False

1. Structured data commonly follows a predefined schema.

Answer: True

2. Unstructured data always fits neatly into relational tables.

Answer: False

3. Median requires ordered data.

Answer: True

4. Mode is the least frequent value.

Answer: False

5. Standard deviation describes dispersion.

Answer: True

D. **Very Short Answer Questions**

1. What is structured data?

Answer: Data organised according to a predefined structure or schema, such as rows and columns in a table.

2. What is unstructured data?

Answer: Data without a fixed tabular schema, such as free-form text, images, audio and video.

3. What is semi-structured data?

Answer: Data that lacks a rigid relational table structure but contains tags, keys or other organisational markers, such as JSON or XML.

4. Define mean.

Answer: Mean is the sum of all values divided by the number of values.

5. Define range.

Answer: Range is the difference between the maximum and minimum values.

E. **Short Answer Questions**

1. Explain the data-processing cycle.

Answer: Data is captured/collected, stored, processed to derive useful results, and retrieved/presented when required. The exact workflow depends on the application.

2. Differentiate structured and unstructured data.

Answer: Structured data has a predefined schema and is easy to organise in tables. Unstructured data lacks a fixed tabular structure and includes media and free-form content.

3. Differentiate mean, median and mode.

Answer: Mean is the arithmetic average; median is the middle ordered value; mode is the most frequent value.

4. Explain data collection.

Answer: Data collection is the systematic gathering of required observations from

appropriate sources so that they can be stored and processed for a defined purpose.

5. What is standard deviation?

Answer: It is the positive square root of the average squared deviation of values from the mean; it indicates how dispersed the data are around the mean.

F. Long Answer / Programming Questions

1. Explain structured, semi-structured and unstructured data with examples.

Answer: Structured data follows a fixed schema such as a student table. Semi-structured data uses tags/keys, such as XML/JSON. Unstructured data has no fixed tabular structure, such as images, audio and free-form text.

2. Explain the complete data-processing cycle.

Answer: Capture or collect relevant data, store it safely, process it using appropriate operations/statistics, and retrieve/present the resulting information for decision making or service delivery.

3. Calculate mean, median, mode and range for 2,3,3,5,7.

Answer: Mean=(20/5)=4; median=3; mode=3; range=7-2=5.

4. Explain why statistical summarisation is useful.

Answer: Large datasets are difficult to inspect individually. Mean, median, mode, range and standard deviation condense important characteristics and help reveal central tendency and dispersion.

5. Describe a scholarship-data processing example.

Answer: Collect marks and family-income data, validate and store records, filter students meeting the stated criteria, process the qualifying records, and retrieve the final beneficiary list.

Additional Questions

1. [MCQ] Which of the following is associated with data arranged in rows and columns in a database?

A) structured data

B) unstructured data

C) audio only

D) random data

Answer: structured data

Explanation: Structured data follows a predefined schema.

2. [MCQ] Which of the following is associated with free-form text, images and videos are common examples of?

A) unstructured data

B) structured data

C) relational keys

D) binary search

Answer: unstructured data

Explanation: Unstructured data does not follow a rigid tabular schema.

3. [MCQ] Which of the following is associated with data such as JSON or XML with tags or keys?

A) semi-structured data

B) structured only

C) unstructured only

D) encrypted data

Answer: semi-structured data

Explanation: Semi-structured data has organizational tags or markers without a rigid relational schema.

4. [MCQ] Which of the following is associated with the first step of many data workflows in the prescribed treatment?

A) data capturing/collection

B) data deletion

C) data sorting only

D) encryption only

Answer: data capturing/collection

Explanation: Data is first captured or collected before storage and processing.

5. [MCQ] Which of the following is associated with the stage where collected data is kept for future use?

A) data storage

B) data retrieval

C) data deletion

D) sorting

Answer: data storage

Explanation: Storage preserves data for later processing and retrieval.

6. [MCQ] Which of the following is associated with the stage that transforms raw data into useful information?

A) data processing

B) data capturing

C) data storage only

D) data disposal

Answer: data processing

Explanation: Processing applies operations to make data meaningful.

7. [MCQ] Which of the following is associated with the stage of obtaining required stored information?

A) data retrieval

B) data capturing

C) data encryption only

D) data deletion

Answer: data retrieval

Explanation: Retrieval accesses needed data from storage.

8. [MCQ] Which of the following is associated with the arithmetic average of values?

A) mean

B) median

C) mode

D) range

Answer: mean

Explanation: Mean is sum of values divided by count.

9. [MCQ] Which of the following is associated with the middle value after ordering a dataset?

A) median

B) mean

C) mode

D) range

Answer: median

Explanation: Median is the central ordered value; for an even count it is the average of the two middle values.

10. [MCQ] Which of the following is associated with the most frequently occurring value?

A) mode

B) mean

C) median

D) range

Answer: mode

Explanation: Mode is the most frequent value.

11. [MCQ] Which of the following is associated with the difference between maximum and minimum values?

A) range

B) mean

C) median

D) mode

Answer: range

Explanation: Range = maximum - minimum.

12. [MCQ] Which of the following is associated with a measure of spread around the mean?

A) standard deviation

B) mode

C) median

D) range only

Answer: standard deviation

Explanation: Standard deviation measures dispersion around the mean.

13. [MCQ] Which of the following is associated with the mean of [2,4,6]?

A) 4

B) 3

C) 5

D) 6

Answer: 4

Explanation: (2+4+6)/3 = 4.

14. [MCQ] Which of the following is associated with the median of [1,3,5,7,9]?

A) 5

B) 3

C) 4

D) 7

Answer: 5

Explanation: 5 is the middle-ordered value.

1. [Fill] Complete the statement: the arithmetic average of values is _____.

Answer: mean

Explanation: Mean is sum of values divided by count.

2. [Fill] Complete the statement: the middle value after ordering a dataset is _____.

Answer: median

Explanation: Median is the central ordered value; for an even count it is the average of the two middle values.

3. [Fill] Complete the statement: the most frequently occurring value is _____.

Answer: mode

Explanation: Mode is the most frequent value.

4. [Fill] Complete the statement: the difference between maximum and minimum values is _____.

Answer: range

Explanation: Range = maximum - minimum.

5. [Fill] Complete the statement: a measure of spread around the mean is _____.

Answer: standard deviation

Explanation: Standard deviation measures dispersion around the mean.

6. [Fill] Complete the statement: the mean of [2,4,6] is _____.

Answer: 4

Explanation: (2+4+6)/3 = 4.

7. [Fill] Complete the statement: the median of [1,3,5,7,9] is _____.

Answer: 5

Explanation: 5 is the middle ordered value.

1. [T/F] True or False: the median of [1,3,5,7,9] is 3.

Answer: False

Explanation: Correct answer: 5. 5 is the middle ordered value.

2. [T/F] True or False: the mode of [2,2,3,4] is 2.

Answer: True

Explanation: 2 occurs most often.

3. [T/F] True or False: the range of [4,9,2] is 6.

Answer: False

Explanation: Correct answer: 7. 9 - 2 = 7.

4. [T/F] True or False: the purpose of data visualization is communicate patterns and insights clearly.

Answer: True

Explanation: Visualization helps people interpret patterns and trends.

5. [T/F] True or False: a dataset with no missing values in every field is described as is encrypted.

Answer: False

Explanation: Correct answer: complete. Completeness means required data values are present.

6. [T/F] True or False: the reason data cleaning is performed is to identify and correct errors or inconsistencies.

Answer: True

Explanation: Cleaning improves data quality before analysis.

1. [VSA] Name the concept associated with: the first step of many data workflows in the prescribed treatment.

Answer: data capturing/collection

Explanation: Data is first captured or collected before storage and processing.

2. [VSA] Name the concept associated with: the stage where collected data is kept for future use.

Answer: data storage

Explanation: Storage preserves data for later processing and retrieval.

3. [VSA] Name the concept associated with: the stage that transforms raw data into useful information.

Answer: data processing

Explanation: Processing applies operations to make data meaningful.

4. [VSA] Name the concept associated with: the stage of obtaining required stored

information.

Answer: data retrieval

Explanation: Retrieval accesses needed data from storage.

5. [VSA] Name the concept associated with: the arithmetic average of values.

Answer: mean

Explanation: Mean is sum of values divided by count.

6. [VSA] Name the concept associated with: the middle value after ordering a dataset.

Answer: median

Explanation: Median is the central ordered value; for an even count it is the average of the two middle values.

1. [Short/Application] Explain the following concept in detail: the stage that transforms raw data into useful information.

Answer: data processing

Explanation: Processing applies operations to make data meaningful.

2. [Short/Application] Explain the following concept in detail: the stage of obtaining required stored information.

Answer: data retrieval

Explanation: Retrieval accesses needed data from storage.

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Explanation: Median is the central ordered value; for an even count it is the average of the two middle values.

5. [Short/Application] Explain the following concept in detail: the most frequently occurring value.

Answer: mode

Explanation: Mode is the most frequent value.

6. [Short/Application] Explain the following concept in detail: the difference between maximum and minimum values.

Answer: range

Explanation: Range = maximum - minimum.

7. [Short/Application] Explain the following concept in detail: a measure of spread around the mean.

Answer: standard deviation

Explanation: Standard deviation measures dispersion around the mean.

LESSON 08 — DATABASE CONCEPTS

A. **Multiple Choice Questions**

1. A DBMS acts as an interface between database and:

A) Users/applications

B) CPU only

C) Keyboard only

D) Printer only

Answer: (A)

2. A row in a relational table is a:

A) Tuple/record

B) Attribute

C) Domain

D) Schema

Answer: (A)

3. A column in a relational table is an:

A) Attribute

B) Tuple

C) Record only

D) Query

Answer: (A)

4. A primary key:

A) Uniquely identifies tuples

B) Stores images only

C) Must be duplicated

D) Is always NULL

Answer: (A)

5. A foreign key is mainly used to:

A) Relate tables

B) Sort rows only

C) Encrypt data

D) Create hardware

Answer: (A)

6. A snapshot of database contents at a time is a database:

A) Instance/state

B) Schema only

C) Engine

D) Key

Answer: (A)

7. The logical structure of a database is its:

A) Schema

B) Instance

C) Query result

D) Row

Answer: (A)

8. A query is a request to a database for:

A) Information/operations

B) Electricity

C) Network topology

D) File compression only

Answer: (A)

9. RDBMS stores data in related:

A) Tables/relations

B) Stacks

C) Queues

D) Packets

Answer: (A)

10. A key made from more than one attribute is a:

A) Composite key

B) Foreign file

C) Simple query

D) Domain

Answer: (A)

B. **Fill in the Blanks**

1. DBMS stands for Database Management _____.

Answer: System

2. A row in a relation is called a _____.

Answer: tuple/record

3. A column is called an _____.

Answer: attribute

4. The unique identifier chosen for a relation is the _____ key.

Answer: primary

5. A key used to relate two tables is a _____ key.

Answer: foreign

C. **True / False**

1. A database can reduce redundancy compared with poorly designed separate files.

Answer: True

2. A primary key should uniquely identify each tuple.

Answer: True

3. A foreign key is used only for sorting.

Answer: False

4. Schema and instance mean exactly the same thing.

Answer: False

5. A query can retrieve information from a database.

Answer: True

D. **Very Short Answer Questions**

1. What is a database?

Answer: A structured collection of related data stored so it can be accessed, managed and updated efficiently.

2. What is DBMS?

Answer: A Database Management System is software used to create, maintain, manipulate and retrieve data from databases.

3. What is a relation?

Answer: In the relational model, a relation is a table consisting of rows and columns.

4. What is a primary key?

Answer: An attribute or set of attributes that uniquely identifies each tuple in a relation.

5. What is a foreign key?

Answer: An attribute or set of attributes that references a key in another relation to establish a relationship.

E. **Short Answer Questions**

1. Explain limitations of a file system.

Answer: Separate files can cause redundancy, inconsistency, difficult sharing, weak centralised control, difficult concurrent access and complicated maintenance.

2. Differentiate schema and instance.

Answer: Schema is the designed structure of a database. Instance/state is the actual contents of the database at a particular time.

3. Differentiate primary and foreign key.

Answer: A primary key uniquely identifies rows in its own table. A foreign key references a key in another table to establish a relationship.

4. Explain degree and cardinality.

Answer: Degree is the number of attributes/columns in a relation. Cardinality is the number of tuples/rows.

5. Explain relational data model.

Answer: It represents data as relations/tables. Rows represent tuples and columns represent attributes, with keys and constraints maintaining relationships and integrity.

F. **Long Answer / Programming Questions**

1. Explain DBMS and its advantages.

Answer: DBMS provides controlled storage, retrieval and manipulation of data. It can reduce redundancy, improve consistency and sharing, enforce integrity and support security and concurrent access.

2. Explain keys in an RDBMS.

Answer: Candidate keys are possible unique identifiers. A primary key is the selected candidate key. A composite key uses multiple attributes. A foreign key references a key in another table.

3. Explain schema, instance and query.

Answer: Schema describes the database structure; instance is its current data state; a query requests retrieval or manipulation of data.

4. Explain conversion from file system to relational database.

Answer: Identify common fields, separate entities into appropriate tables, avoid repeated data, select keys and establish relationships using foreign keys.

5. Discuss DBMS limitations.

Answer: DBMS can increase system complexity, require resources and introduce centralised-data vulnerability; administration, security and maintenance also require careful management.

Additional Questions

1. [MCQ] Which of the following is associated with software that manages databases?

A) DBMS

B) compiler

C) browser

D) operating system only

Answer: DBMS

Explanation: DBMS provides facilities to create, store and manage data.

2. [MCQ] Which of the following is associated with a relational database management system?

A) RDBMS

B) DNS

C) FTP

D) CPU

Answer: RDBMS

Explanation: RDBMS manages data using relations/tables.

3. [MCQ] Which of the following is associated with a row in a relational table?

A) tuple/record

B) attribute

C) domain

D) schema

Answer: tuple/record

Explanation: A tuple represents one record in a relation.

4. [MCQ] Which of the following is associated with a column in a relational table?

A) attribute/field

B) tuple

C) instance

D) database engine

Answer: attribute/field

Explanation: An attribute represents a property of the relation.

5. [MCQ] Which of the following is associated with the set of allowed values for an attribute?

A) domain

B) degree

C) cardinality

D) schema

Answer: domain

Explanation: A domain defines permitted values for an attribute.

6. [MCQ] Which of the following is associated with the number of attributes in a relation?

A) degree

B) cardinality

C) domain

D) instance

Answer: degree

Explanation: Degree counts columns/attributes.

7. [MCQ] Which of the following is associated with the number of tuples in a relation?

A) cardinality

B) degree

C) domain

D) schema

Answer: cardinality

Explanation: Cardinality counts rows/tuples.

8. [MCQ] Which of the following is associated with the key that uniquely identifies each

tuple?

A) primary key

B) foreign key

C) duplicate key

D) domain key

Answer: primary key

Explanation: A primary key provides unique identification.

9. [MCQ] Which of the following is associated with a key that references a key in another relation?

A) foreign key

B) primary key

C) candidate only

D) composite only

Answer: foreign key

Explanation: A foreign key establishes a relationship between tables.

10. [MCQ] Which of the following is associated with a minimal attribute set that can uniquely identify a tuple?

A) candidate key

B) foreign key

C) domain

D) schema

Answer: candidate key

Explanation: Candidate keys are possible choices for the primary key.

11. [MCQ] Which of the following is associated with a key made from two or more attributes?

A) composite key

B) foreign key only

C) single key

D) domain

Answer: composite key

Explanation: A composite key combines multiple attributes.

12. [MCQ] Which of the following is associated with the database description that defines structure?

A) schema

B) instance

C) tuple

D) record

Answer: schema

Explanation: Schema describes the structure of database objects.

13. [MCQ] Which of the following is associated with the data stored in the database at a particular time?

A) instance

B) schema

C) domain

D) degree

Answer: instance

Explanation: An instance is the current state of the database.

14. [MCQ] Which of the following is associated with metadata describing data?

A) data dictionary/metadata

B) tuple

C) primary key

D) query result

Answer: data dictionary/metadata

Explanation: Metadata is information about data and database structure.

15. [MCQ] Which of the following is associated with a special value representing unknown or missing data?

A) NULL

B) 0 always

C) False always

D) empty string always

Answer: NULL

Explanation: NULL represents absence/unknown value, not necessarily zero.

1. [Fill] Complete the statement: the key that uniquely identifies each tuple is _____.

Answer: primary key

Explanation: A primary key provides unique identification.

2. [Fill] Complete the statement: a key that references a key in another relation is _____.

Answer: foreign key

Explanation: A foreign key establishes a relationship between tables.

3. [Fill] Complete the statement: a minimal attribute set that can uniquely identify a tuple is _____.

Answer: candidate key

Explanation: Candidate keys are possible choices for the primary key.

4. [Fill] Complete the statement: a key made from two or more attributes is _____.

Answer: composite key

Explanation: A composite key combines multiple attributes.

5. [Fill] Complete the statement: the database description that defines structure is _____.

Answer: schema

Explanation: Schema describes the structure of database objects.

6. [Fill] Complete the statement: the data stored in the database at a particular time is _____.

Answer: instance

Explanation: An instance is the current state of the database.

7. [Fill] Complete the statement: metadata describing data is _____.

Answer: data dictionary/metadata

Explanation: Metadata is information about data and database structure.

8. [Fill] Complete the statement: a special value representing unknown or missing data is _____.

Answer: NULL

Explanation: NULL represents absence/unknown value, not necessarily zero.

1. [T/F] True or False: metadata describing data is tuple.

Answer: False

Explanation: Correct answer: data dictionary/metadata. Metadata is information about data and database structure.

2. [T/F] True or False: a special value representing unknown or missing data is NULL.

Answer: True

Explanation: NULL represents absence/unknown value, not necessarily zero.

3. [T/F] True or False: a rule that restricts values in a table is query.

Answer: False

Explanation: Correct answer: constraint. Constraints help maintain integrity.

4. [T/F] True or False: the model based on relations/tables is relational data model.

Answer: True

Explanation: The relational model represents data in tables.

5. [T/F] True or False: a request for information from a database is schema.

Answer: False

Explanation: Correct answer: query. A query asks the database to return or manipulate information.

6. [T/F] True or False: a system that creates, manipulates and maintains a relational database is RDBMS.

Answer: True

Explanation: An RDBMS manages relational data and operations.

1. [VSA] Name the concept associated with: a column in a relational table.

Answer: attribute/field

Explanation: An attribute represents a property of the relation.

2. [VSA] Name the concept associated with: the set of allowed values for an attribute.

Answer: domain

Explanation: A domain defines permitted values for an attribute.

3. [VSA] Name the concept associated with: the number of attributes in a relation.

Answer: degree

Explanation: Degree counts columns/attributes.

4. [VSA] Name the concept associated with: the number of tuples in a relation.

Answer: cardinality

Explanation: Cardinality counts rows/tuples.

5. [VSA] Name the concept associated with: the key that uniquely identifies each tuple.

Answer: primary key

Explanation: A primary key provides unique identification.

6. [VSA] Name the concept associated with: a key that references a key in another relation.

Answer: foreign key

Explanation: A foreign key establishes a relationship between tables.

7. [VSA] Name the concept associated with: a minimal attribute set that can uniquely identify a tuple.

Answer: candidate key

Explanation: Candidate keys are possible choices for the primary key.

1. [Short/Application] Explain the following concept in detail: the number of attributes in a relation.

Answer: degree

Explanation: Degree counts columns/attributes.

2. [Short/Application] Explain the following concept in detail: the number of tuples in a relation.

Answer: cardinality

Explanation: Cardinality counts rows/tuples.

3. [Short/Application] Explain the following concept in detail: the key that uniquely identifies each tuple.

Answer: primary key

Explanation: A primary key provides unique identification.

4. [Short/Application] Explain the following concept in detail: a key that references a key in another relation.

Answer: foreign key

Explanation: A foreign key establishes a relationship between tables.

5. [Short/Application] Explain the following concept in detail: a minimal attribute set that can uniquely identify a tuple.

Answer: candidate key

Explanation: Candidate keys are possible choices for the primary key.

6. [Short/Application] Explain the following concept in detail: a key made from two or more attributes.

Answer: composite key

Explanation: A composite key combines multiple attributes.

7. [Short/Application] Explain the following concept in detail: the database description that defines structure.

Answer: schema

Explanation: Schema describes the structure of database objects.

8. [Short/Application] Explain the following concept in detail: the data stored in the database at a particular time.

Answer: instance

Explanation: An instance is the current state of the database.

9. [Short/Application] Explain the following concept in detail: metadata describing data.

Answer: data dictionary/metadata

Explanation: Metadata is information about data and database structure.

LESSON 09 — STRUCTURED QUERY LANGUAGE

A. **Multiple Choice Questions**

1. SQL stands for:

A) Structured Query Language

B) Simple Query Logic

C) System Query Language

D) Structured Queue Language

Answer: (A)

2. Which command creates a database?

A) CREATE DATABASE

B) MAKE DATABASE

C) NEW DATABASE

D) BUILD DATABASE

Answer: (A)

3. Which command creates a table?

A) CREATE TABLE

B) MAKE TABLE

C) NEW TABLE

D) BUILD TABLE

Answer: (A)

4. Which command adds rows?

A) INSERT

B) UPDATE

C) DELETE

D) SELECT

Answer: (A)

5. Which command retrieves rows?

A) SELECT

B) INSERT

C) DROP

D) ALTER

Answer: (A)

6. Which clause filters rows?

A) WHERE

B) ORDER BY

C) GROUP BY only

D) DISTINCT only

Answer: (A)

7. Which clause sorts the result?

A) ORDER BY

B) WHERE

C) VALUES

D) INTO

Answer: (A)

8. Which function returns average?

A) AVG()

B) SUMTEXT()

C) MEANONLY()

D) MIDAVG()

Answer: (A)

9. Which function counts rows/values?

A) COUNT()

B) TOTALROWS()

C) NUMBER()

D) ADD()

Answer: (A)

10. Which command removes a table definition?

A) DROP TABLE

B) DELETE TABLE

C) REMOVE ROW

D) CLEAR TABLE

Answer: (A)

B. **Fill in the Blanks**

1. SQL stands for _____ Query Language.

Answer: Structured

2. The command used to retrieve data is _____.

Answer: SELECT

3. The clause used to filter rows is _____.

Answer: WHERE

4. The clause used to sort output is _____.

Answer: ORDER BY

5. The aggregate function for average is _____.

Answer: AVG()

C. **True / False**

1. SQL can define database structures.

Answer: True

2. DELETE and DROP always mean the same thing.

Answer: False

3. WHERE can filter rows.

Answer: True

4. ORDER BY can sort query output.

Answer: True

5. AVG() is an aggregate function.

Answer: True

D. **Very Short Answer Questions**

1. What is SQL?

Answer: SQL is the standard language used with relational database systems to define, manipulate and query data.

2. What is CREATE TABLE used for?

Answer: It defines a new table and specifies its columns, data types and optional constraints.

3. What is SELECT?

Answer: SELECT retrieves data from one or more tables.

4. What is WHERE?

Answer: WHERE specifies a condition used to filter rows.

5. What is a primary-key constraint?

Answer: It enforces unique identification of each row and does not allow NULL values in the primary-key column(s).

E. **Short Answer Questions**

1. Differentiate DELETE and DROP.

Answer: DELETE removes rows from a table, optionally using a WHERE condition. DROP TABLE removes the table definition and its stored rows.

2. Explain DISTINCT and ORDER BY.

Answer: DISTINCT removes duplicate result rows/values from the selected output. ORDER BY arranges the result according to one or more columns.

3. Explain aggregate functions.

Answer: Functions such as COUNT(), SUM(), AVG(), MIN() and MAX() operate on a set of values and return a summary value.

4. Explain UPDATE.

Answer: UPDATE modifies existing rows. A WHERE condition should normally be used to restrict which rows are changed.

5. What is GROUP BY?

Answer: GROUP BY forms groups of rows with common values so aggregate functions can produce summaries for each group.

F. **Long Answer / Programming Questions**

1. Write basic SQL commands for a student database.

Answer: CREATE DATABASE StudentDB; USE StudentDB; CREATE TABLE Student(RollNo INT PRIMARY KEY, Name VARCHAR(30), Marks INT); INSERT INTO Student VALUES (...); SELECT * FROM Student;

2. Explain SELECT with WHERE, DISTINCT and ORDER BY.

Answer: SELECT chooses columns/expressions, WHERE filters rows, DISTINCT removes duplicate output combinations, and ORDER BY sorts the resulting rows.

3. Explain SQL joins.

Answer: A join combines rows from related tables using a matching condition, commonly through primary/foreign-key relationships. INNER JOIN returns matching rows; other join forms can preserve unmatched rows.

4. Explain SQL functions with examples.

Answer: Numeric functions include MOD() and ROUND(); string functions include UPPER(), LOWER(), LENGTH(), SUBSTRING(); date functions include YEAR(), MONTH() and DAY(); aggregate functions include AVG(), SUM(), MIN(), MAX() and COUNT().

5. Differentiate DDL and DML.

Answer: DDL defines or changes database structures, e.g. CREATE, ALTER and DROP. DML manipulates stored data, e.g. INSERT, UPDATE and DELETE.

Additional Questions

1. [MCQ] Which of the following is associated with the SQL command that creates a database?

A) CREATE DATABASE

B) MAKE DATABASE

C) NEW DATABASE

D) BUILD DB

Answer: CREATE DATABASE

Explanation: CREATE DATABASE creates a database.

2. [MCQ] Which of the following is associated with the SQL command that creates a table?

A) CREATE TABLE

B) MAKE TABLE

C) NEW TABLE

D) BUILD TABLE

Answer: CREATE TABLE

Explanation: CREATE TABLE defines a table structure.

3. [MCQ] Which of the following is associated with the SQL command that adds rows?

A) INSERT INTO

B) ADD ROW

C) APPEND SQL

D) PUT INTO

Answer: INSERT INTO

Explanation: INSERT INTO adds records.

4. [MCQ] Which of the following is associated with the SQL command that modifies existing rows?

A) UPDATE

B) ALTER

C) CHANGE ROW

D) MODIFY RECORD ONLY

Answer: UPDATE

Explanation: UPDATE changes existing data.

5. [MCQ] Which of the following is associated with the SQL command that removes selected rows?

A) DELETE

B) DROP

C) TRUNCATE TABLE ONLY

D) REMOVE TABLE

Answer: DELETE

Explanation: DELETE removes rows, optionally using WHERE.

6. [MCQ] Which of the following is associated with the SQL command that removes a table definition and its data?

A) DROP TABLE

B) DELETE TABLE

C) REMOVE ROWS

D) CLEAR TABLE

Answer: DROP TABLE

Explanation: DROP removes the table object.

7. [MCQ] Which of the following is associated with the SQL command that removes all rows while retaining table structure?

A) TRUNCATE TABLE

B) DROP TABLE

C) DELETE DATABASE

D) REMOVE COLUMN

Answer: TRUNCATE TABLE

Explanation: TRUNCATE removes all rows but keeps the table structure.

8. [MCQ] Which of the following is associated with the clause used to filter rows?

A) WHERE

B) ORDER BY

C) GROUP BY

D) SELECT ONLY

Answer: WHERE

Explanation: WHERE specifies row-selection conditions.

9. [MCQ] Which of the following is associated with the clause used to sort query results?

A) ORDER BY

B) WHERE

C) GROUP BY

D) SORT ROW

Answer: ORDER BY

Explanation: ORDER BY sorts the result.

10. [MCQ] Which of the following is associated with the keyword that removes duplicate result rows?

A) DISTINCT

B) UNIQUE ROWS

C) DIFFERENT

D) ONLY

Answer: DISTINCT

Explanation: DISTINCT eliminates duplicate values from the result set.

11. [MCQ] Which of the following is associated with the clause used to group rows with common values?

A) GROUP BY

B) ORDER BY

C) WHERE

D) DISTINCT

Answer: GROUP BY

Explanation: GROUP BY forms groups for aggregation.

12. [MCQ] Which of the following is associated with the clause used to filter grouped results?

A) HAVING

B) WHERE

C) ORDER BY

D) DISTINCT

Answer: HAVING

Explanation: HAVING applies conditions to groups.

13. [MCQ] Which of the following is associated with the aggregate function that counts rows?

A) COUNT()

B) SUM()

C) AVG()

D) MAX()

Answer: COUNT()

Explanation: COUNT returns a count.

14. [MCQ] Which of the following is associated with the aggregate function that calculates an arithmetic average?

A) AVG()

B) SUM()

C) COUNT()

D) MIN()

Answer: AVG()

Explanation: AVG calculates the average of numeric values.

15. [MCQ] Which of the following is associated with the aggregate function that returns the largest value?

A) MAX()

B) MIN()

C) AVG()

D) COUNT()

Answer: MAX()

Explanation: MAX returns the maximum value.

16. [MCQ] Which of the following is associated with the aggregate function that returns the smallest value?

A) MIN()

B) MAX()

C) AVG()

D) SUM()

Answer: MIN()

Explanation: MIN returns the minimum value.

17. [MCQ] Which of the following is associated with the aggregate function that totals numeric values?

A) SUM()

B) COUNT()

C) AVG()

D) MIN()

Answer: SUM()

Explanation: SUM returns the total.

18. [MCQ] Which of the following is associated with the SQL category containing CREATE, ALTER and DROP?

A) DDL

B) DML

C) DCL

D) HTML

Answer: DDL

Explanation: DDL defines database structures.

19. [MCQ] Which of the following is associated with the SQL category containing INSERT, UPDATE and DELETE?

A) DML

B) DDL

C) DCL

D) TCL only

Answer: DML

Explanation: DML manipulates data.

20. [MCQ] Which of the following is associated with the SQL category associated with

permissions such as GRANT and REVOKE?

A) DCL

B) DDL

C) DML

D) DQL only

Answer: DCL

Explanation: DCL controls access privileges.

21. [MCQ] Which of the following is associated with the wildcard representing any sequence of characters in LIKE?

A) %

B) _

C) *

D) ?

Answer: %

Explanation: % matches zero or more characters in MySQL LIKE patterns.

22. [MCQ] Which of the following is associated with the wildcard representing a single character in LIKE?

A) _

B) %

C) ?

D) *

Answer: _

Explanation: _ matches one character.

23. [MCQ] Which of the following is associated with the SQL operator used to test membership in a list?

A) IN

B) LIKE

C) BETWEEN ONLY

D) MEMBER

Answer: IN

Explanation: IN tests whether a value belongs to a specified set.

24. [MCQ] Which of the following is associated with the SQL operator used to test a range?

A) BETWEEN

B) IN

C) LIKE

D) WITHIN ONLY

Answer: BETWEEN

Explanation: BETWEEN tests an inclusive range.

25. [MCQ] Which of the following is associated with the function that returns the year component of a date?

A) YEAR()

B) MONTHNAME()

C) DATEONLY()

D) DAYYEAR()

Answer: YEAR()

Explanation: YEAR extracts the year.

26. [MCQ] Which of the following is associated with the function that returns the month number?

A) MONTH()

B) YEAR()

C) DATE()

D) MON()

Answer: MONTH()

Explanation: MONTH extracts the month number.

1. [Fill] Complete the statement: the clause used to filter rows is _____.

Answer: WHERE

Explanation: WHERE specifies row-selection conditions.

2. [Fill] Complete the statement: the clause used to sort query results is _____.

Answer: ORDER BY

Explanation: ORDER BY sorts the result.

3. [Fill] Complete the statement: the keyword that removes duplicate result rows is _____.

Answer: DISTINCT

Explanation: DISTINCT eliminates duplicate values from the result set.

4. [Fill] Complete the statement: the clause used to group rows with common values is _____.

Answer: GROUP BY

Explanation: GROUP BY forms groups for aggregation.

5. [Fill] Complete the statement: the clause used to filter grouped results is _____.

Answer: HAVING

Explanation: HAVING applies conditions to groups.

6. [Fill] Complete the statement: the aggregate function that counts rows is _____.

Answer: COUNT()

Explanation: COUNT returns a count.

7. [Fill] Complete the statement: the aggregate function that calculates an arithmetic

average is _____.

Answer: AVG()

Explanation: AVG calculates the average of numeric values.

8. [Fill] Complete the statement: the aggregate function that returns the largest value is _____.

Answer: MAX()

Explanation: MAX returns the maximum value.

9. [Fill] Complete the statement: the aggregate function that returns the smallest value is _____.

Answer: MIN()

Explanation: MIN returns the minimum value.

10. [Fill] Complete the statement: the aggregate function that totals numeric values is _____.

Answer: SUM()

Explanation: SUM returns the total.

11. [Fill] Complete the statement: the SQL category containing CREATE, ALTER and DROP is _____.

Answer: DDL

Explanation: DDL defines database structures.

12. [Fill] Complete the statement: the SQL category containing INSERT, UPDATE and DELETE is _____.

Answer: DML

Explanation: DML manipulates data.

13. [Fill] Complete the statement: the SQL category associated with permissions such as GRANT and REVOKE is _____.

Answer: DCL

Explanation: DCL controls access privileges.

14. [Fill] Complete the statement: the wildcard representing any sequence of characters in LIKE is _____.

Answer: %

Explanation: % matches zero or more characters in MySQL LIKE patterns.

1. [T/F] True or False: the aggregate function that calculates an arithmetic average is SUM().

Answer: False

Explanation: Correct answer: AVG(). AVG calculates the average of numeric values.

2. [T/F] True or False: the aggregate function that returns the largest value is MAX().

Answer: True

Explanation: MAX returns the maximum value.

3. [T/F] True or False: the aggregate function that returns the smallest value is MAX().

Answer: False

Explanation: Correct answer: MIN(). MIN returns the minimum value.

4. [T/F] True or False: the aggregate function that totals numeric values is SUM().

Answer: True

Explanation: SUM returns the total.

5. [T/F] True or False: the SQL category containing CREATE, ALTER and DROP is DML.

Answer: False

Explanation: Correct answer: DDL. DDL defines database structures.

6. [T/F] True or False: the SQL category containing INSERT, UPDATE and DELETE is DML.

Answer: True

Explanation: DML manipulates data.

7. [T/F] True or False: the SQL category associated with permissions such as GRANT and REVOKE is DDL.

Answer: False

Explanation: Correct answer: DCL. DCL controls access privileges.

8. [T/F] True or False: the wildcard representing any sequence of characters in LIKE is %.

Answer: True

Explanation: % matches zero or more characters in MySQL LIKE patterns.

9. [T/F] True or False: the wildcard representing a single character in LIKE is %.

Answer: False

Explanation: Correct answer: _. _ matches one character.

10. [T/F] True or False: the SQL operator used to test membership in a list is IN.

Answer: True

Explanation: IN tests whether a value belongs to a specified set.

11. [T/F] True or False: the SQL operator used to test a range is IN.

Answer: False

Explanation: Correct answer: BETWEEN. BETWEEN tests an inclusive range.

1. [VSA] Name the concept associated with: the SQL command that modifies existing rows.

Answer: UPDATE

Explanation: UPDATE changes existing data.

2. [VSA] Name the concept associated with: the SQL command that removes selected rows.

Answer: DELETE

Explanation: DELETE removes rows, optionally using WHERE.

3. [VSA] Name the concept associated with: the SQL command that removes a table

definition and its data.

Answer: DROP TABLE

Explanation: DROP removes the table object.

4. [VSA] Name the concept associated with: the SQL command that removes all rows while retaining table structure.

Answer: TRUNCATE TABLE

Explanation: TRUNCATE removes all rows but keeps the table structure.

5. [VSA] Name the concept associated with: the clause used to filter rows.

Answer: WHERE

Explanation: WHERE specifies row-selection conditions.

6. [VSA] Name the concept associated with: the clause used to sort query results.

Answer: ORDER BY

Explanation: ORDER BY sorts the result.

7. [VSA] Name the concept associated with: the keyword that removes duplicate result rows.

Answer: DISTINCT

Explanation: DISTINCT eliminates duplicate values from the result set.

8. [VSA] Name the concept associated with: the clause used to group rows with common values.

Answer: GROUP BY

Explanation: GROUP BY forms groups for aggregation.

9. [VSA] Name the concept associated with: the clause used to filter grouped results.

Answer: HAVING

Explanation: HAVING applies conditions to groups.

10. [VSA] Name the concept associated with: the aggregate function that counts rows.

Answer: COUNT()

Explanation: COUNT returns a count.

11. [VSA] Name the concept associated with: the aggregate function that calculates an arithmetic average.

Answer: AVG()

Explanation: AVG calculates the average of numeric values.

12. [VSA] Name the concept associated with: the aggregate function that returns the largest value.

Answer: MAX()

Explanation: MAX returns the maximum value.

1. [Short/Application] Explain the following concept in detail: the SQL command that removes a table definition and its data.

Answer: DROP TABLE

Explanation: DROP removes the table object.

2. [Short/Application] Explain the following concept in detail: the SQL command that removes all rows while retaining table structure.

Answer: TRUNCATE TABLE

Explanation: TRUNCATE removes all rows but keeps the table structure.

3. [Short/Application] Explain the following concept in detail: the clause used to filter rows.

Answer: WHERE

Explanation: WHERE specifies row-selection conditions.

4. [Short/Application] Explain the following concept in detail: the clause used to sort query results.

Answer: ORDER BY

Explanation: ORDER BY sorts the result.

5. [Short/Application] Explain the following concept in detail: the keyword that removes duplicate result rows.

Answer: DISTINCT

Explanation: DISTINCT eliminates duplicate values from the result set.

6. [Short/Application] Explain the following concept in detail: the clause used to group rows with common values.

Answer: GROUP BY

Explanation: GROUP BY forms groups for aggregation.

7. [Short/Application] Explain the following concept in detail: the clause used to filter grouped results.

Answer: HAVING

Explanation: HAVING applies conditions to groups.

8. [Short/Application] Explain the following concept in detail: the aggregate function that counts rows.

Answer: COUNT()

Explanation: COUNT returns a count.

9. [Short/Application] Explain the following concept in detail: the aggregate function that calculates an arithmetic average.

Answer: AVG()

Explanation: AVG calculates the average of numeric values.

10. [Short/Application] Explain the following concept in detail: the aggregate function that returns the largest value.

Answer: MAX()

Explanation: MAX returns the maximum value.

11. [Short/Application] Explain the following concept in detail: the aggregate function that returns the smallest value.

Answer: MIN()

Explanation: MIN returns the minimum value.

12. [Short/Application] Explain the following concept in detail: the aggregate function that totals numeric values.

Answer: SUM()

Explanation: SUM returns the total.

LESSON 10 — COMPUTER NETWORKS

A. **Multiple Choice Questions**

1. A computer network connects:

A) Computing devices for communication/resource sharing

B) Only printers

C) Only CPUs

D) Only storage chips

Answer: (A)

2. PAN generally covers:

A) A small personal range

B) A whole country

C) The globe

D) A continent

Answer: (A)

3. LAN stands for:

A) Local Area Network

B) Large Area Network

C) Long Access Network

D) Local Access Node

Answer: (A)

4. WAN stands for:

A) Wide Area Network

B) Wired Access Network

C) Web Area Node

D) Wide Application Network

Answer: (A)

5. Which device forwards packets between networks?

A) Router

B) Keyboard

C) Monitor

D) Printer

Answer: (A)

6. A star topology has a:

A) Central device

B) Single shared backbone only

C) Ring only

D) Fully connected pair structure

Answer: (A)

7. A MAC address is associated with a:

A) Network interface

B) Database table

C) Web page

D) File extension

Answer: (A)

8. An IP address identifies a:

A) Node/interface on a network

B) File type

C) SQL command

D) Sorting pass

Answer: (A)

9. FTP is mainly used for:

A) File transfer

B) Sending email only

C) Web page styling

D) Database creation

Answer: (A)

10. SMTP is associated with:

A) Email transfer

B) File compression

C) Routing only

D) Image editing

Answer: (A)

B. **Fill in the Blanks**

1. A device connected to a network is called a _____.

Answer: node

2. PAN means Personal Area _____.

Answer: Network

3. LAN means Local Area _____.

Answer: Network

4. A _____ forwards packets between networks.

Answer: router

5. The physical/hardware address is called the _____ address.

Answer: MAC

C. **True / False**

1. WAN generally covers a larger geographical area than LAN.

Answer: True

2. A router connects/forwards traffic between networks.

Answer: True

3. Star topology has no central connecting device.

Answer: False

4. MAC and IP addresses serve identical purposes.

Answer: False

5. FTP is used for file transfer.

Answer: True

D. **Very Short Answer Questions**

1. What is a computer network?

Answer: It is an interconnection of two or more computing devices that enables communication and resource sharing.

2. What is a node?

Answer: A node is a device participating in a network and capable of sending, receiving, storing or forwarding data.

3. What is a router?

Answer: A router connects networks and forwards packets along appropriate routing paths.

4. What is star topology?

Answer: A topology in which each communicating device connects to a central device such as a switch or hub.

5. What is an IP address?

Answer: An Internet Protocol address is a logical address used to identify a node/interface on a network.

E. **Short Answer Questions**

1. Differentiate LAN, MAN and WAN.

Answer: LAN covers a local area such as a building/campus. MAN covers a metropolitan area. WAN covers very large geographical regions and interconnects distant networks.

2. Differentiate MAC and IP address.

Answer: MAC is a hardware/physical address associated with a network interface. IP is a

logical network address that can change when the device moves between networks.

3. Explain common network topologies.

Answer: Mesh connects many/all devices directly; ring connects each node to two neighbours; bus uses a shared backbone; star connects nodes to a central device; tree/hybrid combines branching/basic topologies.

4. Explain router and gateway.

Answer: A router forwards packets between networks. A gateway serves as an entry/exit point between networks and can connect networks using different protocols or architectures.

5. Differentiate Internet and WWW.

Answer: The Internet is the global network infrastructure. The World Wide Web is a system of interlinked web pages/resources accessed using Internet technologies.

F. **Long Answer / Programming Questions**

1. Explain PAN, LAN, MAN and WAN.

Answer: PAN connects nearby personal devices. LAN connects devices in a limited local area. MAN covers a city/metropolitan region. WAN covers large geographical areas and links distant networks.

2. Explain star, bus, ring and mesh topology.

Answer: Star uses a central device; bus uses a shared backbone; ring connects nodes in a loop; mesh provides multiple/direct connections between devices. Each has different cost, reliability and maintenance characteristics.

3. Explain networking devices.

Answer: Hub broadcasts data to connected ports; switch forwards frames using learned addresses; router forwards packets between networks; modem performs modulation/demodulation for compatible access links; gateway provides inter-network entry/exit functionality.

4. Explain Internet services/protocols.

Answer: HTTP/HTTPS support web communication, FTP transfers files, SMTP handles outgoing email transfer, and DNS maps domain names to network addresses.

5. Explain MAC and IP addressing.

Answer: MAC provides hardware-level interface identification on a local network. IP provides logical addressing used for routing packets across interconnected networks.

Additional Questions

1. [MCQ] Which of the following is associated with a network covering a small local area such as a building?

A) LAN

B) WAN

C) MAN

D) PANIC

Answer: LAN

Explanation: LAN is a Local Area Network.

2. [MCQ] Which of the following is associated with a network covering a metropolitan area?

A) MAN

B) LAN

C) WAN

D) PAN

Answer: MAN

Explanation: MAN is a Metropolitan Area Network.

3. [MCQ] Which of the following is associated with a network covering a very large geographical area?

A) WAN

B) LAN

C) MAN

D) PAN

Answer: WAN

Explanation: WAN is a Wide Area Network.

4. [MCQ] Which of the following is associated with a network around a person and nearby devices?

A) PAN

B) WAN

C) MAN

D) LAN only

Answer: PAN

Explanation: PAN is a Personal Area Network.

5. [MCQ] Which of the following is associated with the device that connects different networks and forwards packets?

A) router

B) hub

C) repeater

D) keyboard

Answer: router

Explanation: Routers connect networks and route packets.

6. [MCQ] Which of the following is associated with the device that forwards frames within a LAN using addresses?

A) switch

B) router only

C) modem only

D) repeater only

Answer: switch

Explanation: A switch connects LAN devices and forwards frames based on MAC information.

7. [MCQ] Which of the following is associated with the device that regenerates or repeats a signal?

A) repeater

B) router

C) switch

D) gateway only

Answer: repeater

Explanation: A repeater extends a network by regenerating signals.

8. [MCQ] Which of the following is associated with the device that converts digital and analog signals for communication links?

A) modem

B) switch

C) hub

D) gateway only

Answer: modem

Explanation: Modem comes from modulator-demodulator.

9. [MCQ] Which of the following is associated with the topology with a central connecting device?

A) star

B) bus

C) ring

D) mesh

Answer: star

Explanation: Star topology connects nodes to a central hub/switch.

10. [MCQ] Which of the following is associated with the topology using a single backbone cable?

A) bus

B) star

C) ring

D) mesh

Answer: bus

Explanation: Bus topology uses a shared backbone.

11. [MCQ] Which of the following is associated with the topology in which devices form a closed loop?

A) ring

B) star

C) bus

D) tree only

Answer: ring

Explanation: Ring topology connects nodes in a loop.

12. [MCQ] Which of the following is associated with the topology in which nodes have many direct interconnections?

A) mesh

B) bus

C) star

D) ring

Answer: mesh

Explanation: Mesh provides multiple paths between nodes.

13. [MCQ] Which of the following is associated with the protocol used to transfer files?

A) FTP

B) SMTP

C) HTTP only

D) DNS

Answer: FTP

Explanation: FTP is File Transfer Protocol.

14. [MCQ] Which of the following is associated with the protocol used to send email?

A) SMTP

B) FTP

C) HTTP

D) DNS

Answer: SMTP

Explanation: SMTP is used for sending email.

15. [MCQ] Which of the following is associated with the protocol used for web pages?

A) HTTP

B) FTP

C) SMTP

D) DNS

Answer: HTTP

Explanation: HTTP is Hypertext Transfer Protocol.

16. [MCQ] Which of the following is associated with the secure version of HTTP using encryption?

A) HTTPS

B) HTTP

C) FTP

D) SMTP

Answer: HTTPS

Explanation: HTTPS secures HTTP communication using TLS.

17. [MCQ] Which of the following is associated with the system that translates domain names to IP addresses?

A) DNS

B) DHCP

C) FTP

D) SMTP

Answer: DNS

Explanation: DNS resolves names to IP addresses.

18. [MCQ] Which of the following is associated with the hardware address associated with a network interface?

A) MAC address

B) IP address

C) URL

D) DNS name

Answer: MAC address

Explanation: A MAC address identifies a network interface at the link layer.

19. [MCQ] Which of the following is associated with the logical network address used for routing?

A) IP address

B) MAC address

C) URL only

D) domain name only

Answer: IP address

Explanation: IP addresses identify hosts/interfaces for network routing.

1. [Fill] Complete the statement: the device that converts digital and analog signals for communication links is _____.

Answer: modem

Explanation: Modem comes from modulator-demodulator.

2. [Fill] Complete the statement: the topology with a central connecting device is _____.

Answer: star

Explanation: Star topology connects nodes to a central hub/switch.

3. [Fill] Complete the statement: the topology using a single backbone cable is _____.

Answer: bus

Explanation: Bus topology uses a shared backbone.

4. [Fill] Complete the statement: the topology in which devices form a closed loop is _____.

Answer: ring

Explanation: Ring topology connects nodes in a loop.

5. [Fill] Complete the statement: the topology in which nodes have many direct interconnections is _____.

Answer: mesh

Explanation: Mesh provides multiple paths between nodes.

6. [Fill] Complete the statement: the protocol used to transfer files is _____.

Answer: FTP

Explanation: FTP is File Transfer Protocol.

7. [Fill] Complete the statement: the protocol used to send email is _____.

Answer: SMTP

Explanation: SMTP is used for sending email.

8. [Fill] Complete the statement: the protocol used for web pages is _____.

Answer: HTTP

Explanation: HTTP is Hypertext Transfer Protocol.

9. [Fill] Complete the statement: the secure version of HTTP using encryption is _____.

Answer: HTTPS

Explanation: HTTPS secures HTTP communication using TLS.

10. [Fill] Complete the statement: the system that translates domain names to IP addresses is _____.

Answer: DNS

Explanation: DNS resolves names to IP addresses.

1. [T/F] True or False: the protocol used to send email is FTP.

Answer: False

Explanation: Correct answer: SMTP. SMTP is used for sending email.

2. [T/F] True or False: the protocol used for web pages is HTTP.

Answer: True

Explanation: HTTP is Hypertext Transfer Protocol.

3. [T/F] True or False: the secure version of HTTP using encryption is HTTP.

Answer: False

Explanation: Correct answer: HTTPS. HTTPS secures HTTP communication using TLS.

4. [T/F] True or False: the system that translates domain names to IP addresses is DNS.

Answer: True

Explanation: DNS resolves names to IP addresses.

5. [T/F] True or False: the hardware address associated with a network interface is IP address.

Answer: False

Explanation: Correct answer: MAC address. A MAC address identifies a network interface at the link layer.

6. [T/F] True or False: the logical network address used for routing is IP address.

Answer: True

Explanation: IP addresses identify hosts/interfaces for network routing.

7. [T/F] True or False: the standard IPv4 address length is 64 bits.

Answer: False

Explanation: Correct answer: 32 bits. IPv4 uses 32-bit addresses.

8. [T/F] True or False: the standard IPv6 address length is 128 bits.

Answer: True

Explanation: IPv6 uses 128-bit addresses.

1. [VSA] Name the concept associated with: a network around a person and nearby devices.

Answer: PAN

Explanation: PAN is a Personal Area Network.

2. [VSA] Name the concept associated with: the device that connects different networks and forwards packets.

Answer: router

Explanation: Routers connect networks and route packets.

3. [VSA] Name the concept associated with: the device that forwards frames within a LAN using addresses.

Answer: switch

Explanation: A switch connects LAN devices and forwards frames based on MAC information.

4. [VSA] Name the concept associated with: the device that regenerates or repeats a signal.

Answer: repeater

Explanation: A repeater extends a network by regenerating signals.

5. [VSA] Name the concept associated with: the device that converts digital and analog signals for communication links.

Answer: modem

Explanation: Modem comes from modulator-demodulator.

6. [VSA] Name the concept associated with: the topology with a central connecting device.

Answer: star

Explanation: Star topology connects nodes to a central hub/switch.

7. [VSA] Name the concept associated with: the topology using a single backbone cable.

Answer: bus

Explanation: Bus topology uses a shared backbone.

8. [VSA] Name the concept associated with: the topology in which devices form a closed loop.

Answer: ring

Explanation: Ring topology connects nodes in a loop.

9. [VSA] Name the concept associated with: the topology in which nodes have many direct interconnections.

Answer: mesh

Explanation: Mesh provides multiple paths between nodes.

1. [Short/Application] Explain the following concept in detail: the device that forwards frames within a LAN using addresses.

Answer: switch

Explanation: A switch connects LAN devices and forwards frames based on MAC information.

2. [Short/Application] Explain the following concept in detail: the device that regenerates or repeats a signal.

Answer: repeater

Explanation: A repeater extends a network by regenerating signals.

3. [Short/Application] Explain the following concept in detail: the device that converts digital and analog signals for communication links.

Answer: modem

Explanation: Modem comes from modulator-demodulator.

4. [Short/Application] Explain the following concept in detail: the topology with a central connecting device.

Answer: star

Explanation: Star topology connects nodes to a central hub/switch.

5. [Short/Application] Explain the following concept in detail: the topology using a single backbone cable.

Answer: bus

Explanation: Bus topology uses a shared backbone.

6. [Short/Application] Explain the following concept in detail: the topology in which devices form a closed loop.

Answer: ring

Explanation: Ring topology connects nodes in a loop.

7. [Short/Application] Explain the following concept in detail: the topology in which nodes have many direct interconnections.

Answer: mesh

Explanation: Mesh provides multiple paths between nodes.

8. [Short/Application] Explain the following concept in detail: the protocol used to transfer files.

Answer: FTP

Explanation: FTP is File Transfer Protocol.

9. [Short/Application] Explain the following concept in detail: the protocol used to send email.

Answer: SMTP

Explanation: SMTP is used for sending email.

LESSON 11 — DATA COMMUNICATION

A. **Multiple Choice Questions**

1. Data communication means:

A) Exchange of data between connected devices

B) Only data storage

C) Only sorting

D) Only encryption

Answer: (A)

2. Which is NOT one of the five basic components?

A) Sender

B) Receiver

C) Message

D) Compiler

Answer: (D)

3. Bandwidth refers to:

A) Capacity/range of a communication channel

B) File name

C) CPU speed only

D) Password length

Answer: (A)

4. In simplex communication, data flows:

A) One direction only

B) Both directions simultaneously

C) Both directions alternately

D) Randomly

Answer: (A)

5. Full duplex allows:

A) Both directions simultaneously

B) One direction only

C) No communication

D) Only broadcast

Answer: (A)

6. Which is guided medium?

A) Optical fibre

B) Radio wave

C) Microwave

D) Infrared through air

Answer: (A)

7. Which is unguided?

A) Radio wave

B) Twisted pair

C) Coaxial cable

D) Optical fibre

Answer: (A)

8. FTP is used for:

A) File transfer

B) Email sending

C) Web styling

D) Encryption

Answer: (A)

9. SMTP is used for:

A) Email transfer

B) File storage

C) Sorting

D) Database joins

Answer: (A)

10. Optical fibre transmits data using:

A) Light

B) Sound

C) Magnetic ink

D) Mechanical motion

Answer: (A)

B. **Fill in the Blanks**

1. The five components include sender, receiver, message, medium and _____.

Answer: protocol

2. The capacity of a communication channel is related to its _____.

Answer: bandwidth

3. One-way communication is called _____.

Answer: simplex

4. Communication in both directions at the same time is _____ duplex.

Answer: full

5. Optical fibre uses _____ signals.

Answer: light

C. **True / False**

1. A protocol defines rules for communication.

Answer: True

2. Simplex permits simultaneous two-way communication.

Answer: False

3. Twisted pair is a guided medium.

Answer: True

4. Radio waves are unguided.

Answer: True

5. Bandwidth and data transfer rate are always identical concepts.

Answer: False

D. **Very Short Answer Questions**

1. Define data communication.

Answer: It is the exchange of data between two or more networked/connected devices through a communication medium.

2. Name the five components.

Answer: Sender, receiver, message, transmission medium and protocol.

3. What is bandwidth?

Answer: Bandwidth describes the capacity of a communication channel to carry data, often expressed in bits per second for digital links.

4. What is simplex communication?

Answer: Communication in which data travels in only one direction.

5. Name two guided media.

Answer: Twisted-pair cable and optical fibre are guided transmission media.

E. Short Answer Questions

1. Differentiate simplex, half duplex and full duplex.

Answer: Simplex is one-way. Half duplex permits two-way communication but not simultaneously. Full duplex permits simultaneous two-way communication.

2. Differentiate bandwidth and data transfer rate.

Answer: Bandwidth describes channel capacity, while data transfer rate is the actual rate at which data is transmitted over a link under given conditions.

3. Compare twisted pair, coaxial and optical fibre.

Answer: Twisted pair is inexpensive and common for short links; coaxial offers better shielding; optical fibre provides high bandwidth, long distance and immunity to electromagnetic interference.

4. What is a protocol and why is it needed?

Answer: A protocol is a set of communication rules. It ensures devices agree on formats, timing, addressing, error handling and other procedures.

5. Explain switching.

Answer: Switching determines how data is moved through a network from source to destination. Common approaches include circuit switching, packet switching and message switching.

F. **Long Answer / Programming Questions**

1. Explain the components of data communication.

Answer: A sender originates data; the receiver accepts it; the message is the information; the transmission medium carries it; and the protocol defines communication rules.

2. Explain guided and unguided media.

Answer: Guided media carry signals through a physical path such as twisted pair, coaxial or optical fibre. Unguided media use electromagnetic waves through space, such as radio, microwave and infrared.

3. Explain radio, microwave and infrared.

Answer: Radio waves can cover broad areas and support wireless communication. Microwaves offer directional, high-frequency communication and are used in links/satellites. Infrared is short-range and generally line-of-sight, commonly used for nearby devices.

4. Explain TCP/IP and SMTP.

Answer: TCP/IP is a suite of networking protocols used for Internet communication; TCP provides reliable transport while IP handles addressing/routing. SMTP is an application-layer protocol used for email transfer.

5. Explain switching techniques.

Answer: Circuit switching reserves a communication path; packet switching divides data into packets that may share network paths; message switching stores and forwards complete messages at intermediate nodes.

Additional Questions

1. [MCQ] Which of the following is associated with the component that originates a message?

A) sender

B) receiver

C) medium

D) protocol only

Answer: sender

Explanation: The sender initiates data communication.

2. [MCQ] Which of the following is associated with the component that accepts a message?

A) receiver

B) sender

C) router only

D) protocol only

Answer: receiver

Explanation: The receiver is the destination of the message.

3. [MCQ] Which of the following is associated with the path used to carry a signal?

A) transmission medium

B) sender

C) receiver

D) data source only

Answer: transmission medium

Explanation: A medium carries the signal from sender to receiver.

4. [MCQ] Which of the following is associated with the rules governing communication?

A) protocol

B) bandwidth

C) frequency

D) storage

Answer: protocol

Explanation: Protocols define how communicating systems exchange data.

5. [MCQ] Which of the following is associated with the information being transmitted?

A) message/data

B) protocol

C) medium

D) frequency only

Answer: message/data

Explanation: The message is the data being communicated.

6. [MCQ] Which of the following is associated with the unit commonly used for data transfer rate?

A) bits per second (bps)

B) bytes per minute only

C) hertz only

D) volts

Answer: bits per second (bps)

Explanation: Data transfer rate is commonly measured in bps.

7. [MCQ] Which of the following is associated with the mode in which communication is one-way?

A) simplex

B) half duplex

C) full duplex

D) multiplex

Answer: simplex

Explanation: Simplex permits transmission in one direction.

8. [MCQ] Which of the following is associated with the mode in which communication is two-way but not simultaneous?

A) half duplex

B) simplex

C) full duplex

D) multiplex

Answer: half duplex

Explanation: Half duplex allows either direction, one at a time.

9. [MCQ] Which of the following is associated with the mode in which both directions can operate simultaneously?

A) full duplex

B) simplex

C) half duplex

D) broadcast only

Answer: full duplex

Explanation: Full duplex supports simultaneous two-way communication.

10. [MCQ] Which of the following is associated with the guided medium made from twisted copper pairs?

A) twisted pair cable

B) radio waves

C) infrared

D) microwave

Answer: twisted pair cable

Explanation: Twisted pair is a guided wired medium.

11. [MCQ] Which of the following is associated with the guided medium with a central conductor and shield?

A) coaxial cable

B) optical fibre

C) radio waves

D) infrared

Answer: coaxial cable

Explanation: Coaxial cable uses a central conductor and shielding.

12. [MCQ] Which of the following is associated with the guided medium carrying light signals?

A) optical fibre

B) twisted pair

C) coaxial cable

D) radio wave

Answer: optical fibre

Explanation: Optical fibre transmits data using light.

1. [Fill] Complete the statement: the mode in which communication is two-way but not simultaneous is _____.

Answer: half duplex

Explanation: Half duplex allows either direction, one at a time.

2. [Fill] Complete the statement: the mode in which both directions can operate simultaneously is _____.

Answer: full duplex

Explanation: Full duplex supports simultaneous two-way communication.

3. [Fill] Complete the statement: the guided medium made from twisted copper pairs is _____.

Answer: twisted pair cable

Explanation: Twisted pair is a guided wired medium.

4. [Fill] Complete the statement: the guided medium with a central conductor and shield is _____.

Answer: coaxial cable

Explanation: Coaxial cable uses a central conductor and shielding.

5. [Fill] Complete the statement: the guided medium carrying light signals is _____.

Answer: optical fibre

Explanation: Optical fibre transmits data using light.

6. [Fill] Complete the statement: an unguided medium using electromagnetic radio signals is _____.

Answer: radio waves

Explanation: Radio waves propagate through space.

1. [T/F] True or False: an unguided medium commonly requiring line-of-sight antennas is twisted pair.

Answer: False

Explanation: Correct answer: microwaves. Microwave links often use directional line-of-sight paths.

2. [T/F] True or False: an electromagnetic medium commonly used for short-range remote control is infrared.

Answer: True

Explanation: Infrared is used for short-range communication and remote controls.

3. [T/F] True or False: the technique that divides communication into paths/resources for transfer is sorting.

Answer: False

Explanation: Correct answer: switching. Switching selects a path or establishes a communication route.

4. [T/F] True or False: the technique where a dedicated path is reserved for the duration is circuit switching.

Answer: True

Explanation: Circuit switching establishes a dedicated communication path.

5. [T/F] True or False: the technique that sends data as packets is circuit switching.

Answer: False

Explanation: Correct answer: packet switching. Packet switching divides data into packets.

1. [VSA] Name the concept associated with: the rules governing communication.

Answer: protocol

Explanation: Protocols define how communicating systems exchange data.

2. [VSA] Name the concept associated with: the information being transmitted.

Answer: message/data

Explanation: The message is the data being communicated.

3. [VSA] Name the concept associated with: the unit commonly used for data transfer rate.

Answer: bits per second (bps)

Explanation: Data transfer rate is commonly measured in bps.

4. [VSA] Name the concept associated with: the mode in which communication is one-way.

Answer: simplex

Explanation: Simplex permits transmission in one direction.

5. [VSA] Name the concept associated with: the mode in which communication is two-way but not simultaneous.

Answer: half duplex

Explanation: Half duplex allows either direction, one at a time.

6. [VSA] Name the concept associated with: the mode in which both directions can

operate simultaneously.

Answer: full duplex

Explanation: Full duplex supports simultaneous two-way communication.

1. [Short/Application] Explain the following concept in detail: the unit commonly used for data transfer rate.

Answer: bits per second (bps)

Explanation: Data transfer rate is commonly measured in bps.

2. [Short/Application] Explain the following concept in detail: the mode in which communication is one-way.

Answer: simplex

Explanation: Simplex permits transmission in one direction.

3. [Short/Application] Explain the following concept in detail: the mode in which communication is two-way but not simultaneous.

Answer: half duplex

Explanation: Half duplex allows either direction, one at a time.

4. [Short/Application] Explain the following concept in detail: the mode in which both directions can operate simultaneously.

Answer: full duplex

Explanation: Full duplex supports simultaneous two-way communication.

5. [Short/Application] Explain the following concept in detail: the guided medium made from twisted copper pairs.

Answer: twisted pair cable

Explanation: Twisted pair is a guided wired medium.

6. [Short/Application] Explain the following concept in detail: the guided medium with a central conductor and shield.

Answer: coaxial cable

Explanation: Coaxial cable uses a central conductor and shielding.

LESSON 12 — SECURITY ASPECTS

A. **Multiple Choice Questions**

1. Which is malware?

A) Virus

B) Compiler

C) Router

D) Spreadsheet

Answer: (A)

2. A Trojan typically:

A) Disguises itself as legitimate software

B) Is a network cable

C) Sorts data

D) Creates tables

Answer: (A)

3. A worm can:

A) Spread automatically across networks

B) Only display images

C) Act only as a printer

D) Create SQL tables

Answer: (A)

4. Spyware is designed to:

A) Secretly monitor/collect information

B) Increase bandwidth

C) Sort records

D) Create backups only

Answer: (A)

5. Ransomware commonly:

A) Restricts/encrypts access and demands payment

B) Improves security

C) Speeds networks

D) Creates passwords

Answer: (A)

6. A firewall helps:

A) Control network traffic and unauthorised access

B) Sort arrays

C) Create SQL tables

D) Compress images only

Answer: (A)

7. HTTPS adds security to HTTP mainly through:

A) Encryption/TLS protection

B) Sorting

C) Compression only

D) Printing

Answer: (A)

8. DoS means:

A) Denial of Service

B) Data on Server

C) Domain Operating System

D) Digital Output Service

Answer: (A)

9. DDoS uses:

A) Multiple sources to attack a target service

B) One keyboard

C) One file only

D) Only encryption

Answer: (A)

10. Secret interception of private communication is:

A) Eavesdropping

B) Sorting

C) Indexing

D) Parsing

Answer: (A)

B. **Fill in the Blanks**

1. A program that can replicate and spread is a computer _____.

Answer: virus

2. A Trojan disguises itself as _____ software.

Answer: legitimate/useful

3. The process of converting plain text into cipher text is called _____.

Answer: encryption

4. A _____ controls incoming and outgoing network traffic according to rules.

Answer: firewall

5. HTTPS is the secure form of _____ for web communication.

Answer: HTTP

C. **True / False**

1. A worm can spread without attaching to another program in the same way a traditional virus does.

Answer: True

2. A firewall and antivirus perform exactly the same function.

Answer: False

3. HTTPS provides stronger protection than plain HTTP against interception when correctly configured.

Answer: True

4. DDoS commonly involves many compromised systems.

Answer: True

5. Snooping and eavesdropping are unrelated to network security.

Answer: False

D. **Very Short Answer Questions**

1. What is malware?

Answer: Malware is malicious software designed to disrupt, damage, misuse or gain unauthorised access to systems or data.

2. What is a computer virus?

Answer: A virus is malicious code that can replicate by attaching to files/programs and may spread when the infected host is executed or shared.

3. What is ransomware?

Answer: Ransomware is malware that restricts access to data or systems, often by encryption, and demands payment.

4. What is a firewall?

Answer: A firewall is a security control that filters network traffic according to configured rules.

5. What is phishing?

Answer: Phishing is a social-engineering technique in which attackers impersonate trusted entities to trick users into revealing information or performing unsafe actions.

E. **Short Answer Questions**

1. Differentiate virus, worm and Trojan.

Answer: A virus commonly attaches to a host file/program; a worm can self-propagate across systems/networks; a Trojan disguises malicious functionality as legitimate software and relies on the user running/installing it.

2. Differentiate antivirus and firewall.

Answer: Antivirus detects/prevents/removes malicious software on a system. A firewall controls network traffic between trust zones according to rules. They address different layers of protection.

3. Explain HTTP vs HTTPS.

Answer: HTTP transfers web data without the same transport encryption protections. HTTPS uses TLS to provide encrypted communication and server authentication, reducing interception/tampering risks.

4. Differentiate DoS and DDoS.

Answer: DoS attempts to make a service unavailable, often from one source. DDoS distributes the attack across many systems, making mitigation more difficult.

5. Differentiate snooping and eavesdropping.

Answer: Snooping involves secret capture/analysis of network traffic. Eavesdropping specifically refers to unauthorised real-time interception or monitoring of private communication.

F. **Long Answer / Programming Questions**

1. Explain malware distribution and prevention.

Answer: Malware may spread through malicious downloads, infected attachments, compromised websites, removable media, unsafe links or vulnerable software. Prevention includes updates, trusted sources, antivirus, firewalls, backups, least privilege and user awareness.

2. Explain antivirus identification methods.

Answer: Antivirus products can use signatures to recognise known malware patterns, heuristic/behavioural techniques to detect suspicious activity, and other analysis methods to identify previously unseen threats.

3. Explain firewall working.

Answer: A firewall examines traffic against configured rules based on attributes such as source/destination, ports, protocols and connection state, then permits, blocks or logs traffic accordingly.

4. Explain cookies and their security/privacy implications.

Answer: Cookies store information associated with web sessions or preferences. They can improve convenience and maintain sessions, but misuse or theft can create privacy and session-security risks; secure flags and appropriate policies reduce exposure.

5. Explain hackers, DoS/DDoS, snooping and eavesdropping.

Answer: Hackers are people who access or manipulate systems; the security impact depends on intent and authorization. DoS/DDoS target availability. Snooping captures/analyses traffic secretly, while eavesdropping monitors private communication without authorization.

Additional Questions

1. [MCQ] Which of the following is associated with software designed to harm, disrupt, or gain unauthorized access?

A) malware

B) middleware

C) firmware

D) freeware

Answer: malware

Explanation: Malware is malicious software.

2. [MCQ] Which of the following is associated with malware that attaches itself to files/programs and replicates when they are executed?

A) virus

B) worm

C) trojan

D) firewall

Answer: virus

Explanation: A virus typically needs a host file/program for replication.

3. [MCQ] Which of the following is associated with malware that can self-propagate

across networks?

A) worm

B) virus

C) trojan

D) adware

Answer: worm

Explanation: A worm can spread without attaching to a host file in the same way as a virus.

4. [MCQ] Which of the following is associated with malware that disguises itself as legitimate software?

A) Trojan horse

B) worm

C) firewall

D) antivirus

Answer: Trojan horse

Explanation: A Trojan masquerades as legitimate or useful software.

5. [MCQ] Which of the following is associated with software that secretly monitors user activity?

A) spyware

B) antivirus

C) firewall

D) compiler

Answer: spyware

Explanation: Spyware collects information without appropriate user awareness/consent.

6. [MCQ] Which of the following is associated with software that displays unwanted advertisements?

A) adware

B) firewall

C) worm

D) backup

Answer: adware

Explanation: Adware is associated with unwanted advertising.

7. [MCQ] Which of the following is associated with malware that encrypts data and demands payment?

A) ransomware

B) spyware

C) adware

D) firewall

Answer: ransomware

Explanation: Ransomware commonly denies access to data and demands a ransom.

8. [MCQ] Which of the following is associated with software used to detect and remove known malicious software?

A) antivirus

B) compiler

C) router

D) browser only

Answer: antivirus

Explanation: Antivirus software detects, blocks, quarantines or removes malware.

9. [MCQ] Which of the following is associated with a security system controlling incoming and outgoing network traffic?

A) firewall

B) switch

C) hub

D) compiler

Answer: firewall

Explanation: A firewall applies traffic-control rules.

10. [MCQ] Which of the following is associated with the process of converting readable plaintext to ciphertext?

A) encryption

B) decryption

C) compression

D) sorting

Answer: encryption

Explanation: Encryption transforms plaintext into protected ciphertext.

11. [MCQ] Which of the following is associated with the process of converting ciphertext back to plaintext?

A) decryption

B) encryption

C) hashing only

D) compression

Answer: decryption

Explanation: Decryption reverses encryption when the required key/process is available.

12. [MCQ] Which of the following is associated with a fraudulent attempt to obtain sensitive information by pretending to be trustworthy?

A) phishing

B) sorting

C) backup

D) compression

Answer: phishing

Explanation: Phishing uses deceptive messages or sites to obtain information.

13. [MCQ] Which of the following is associated with secretly observing data or communications?

A) snooping/eavesdropping

B) encryption

C) backup

D) authentication

Answer: snooping/eavesdropping

Explanation: Snooping or eavesdropping involves unauthorized observation.

14. [MCQ] Which of the following is associated with an attack that overwhelms a service with traffic or requests?

A) DoS/DDoS

B) antivirus

C) encryption

D) backup

Answer: DoS/DDoS

Explanation: Denial-of-service attacks seek to make a service unavailable.

1. [Fill] Complete the statement: software used to detect and remove known malicious software is _____.

Answer: antivirus

Explanation: Antivirus software detects, blocks, quarantines or removes malware.

2. [Fill] Complete the statement: a security system controlling incoming and outgoing network traffic is _____.

Answer: firewall

Explanation: A firewall applies traffic-control rules.

3. [Fill] Complete the statement: the process of converting readable plaintext to ciphertext is _____.

Answer: encryption

Explanation: Encryption transforms plaintext into protected ciphertext.

4. [Fill] Complete the statement: the process of converting ciphertext back to plaintext is _____.

Answer: decryption

Explanation: Decryption reverses encryption when the required key/process is available.

5. [Fill] Complete the statement: a fraudulent attempt to obtain sensitive information by pretending to be trustworthy is _____.

Answer: phishing

Explanation: Phishing uses deceptive messages or sites to obtain information.

6. [Fill] Complete the statement: secretly observing data or communications is _____.

Answer: snooping/eavesdropping

Explanation: Snooping or eavesdropping involves unauthorized observation.

7. [Fill] Complete the statement: an attack that overwhelms a service with traffic or requests is _____.

Answer: DoS/DDoS

Explanation: Denial-of-service attacks seek to make a service unavailable.

1. [T/F] True or False: an attack that overwhelms a service with traffic or requests is antivirus.

Answer: False

Explanation: Correct answer: DoS/DDoS. Denial-of-service attacks seek to make a service unavailable.

2. [T/F] True or False: a mechanism verifying a user identity is authentication.

Answer: True

Explanation: Authentication establishes who the user is.

3. [T/F] True or False: a mechanism determining what an authenticated user may access is authentication.

Answer: False

Explanation: Correct answer: authorization. Authorization defines permitted actions/resources.

4. [T/F] True or False: a small piece of web data stored by a browser/site is cookie.

Answer: True

Explanation: Cookies can store session or preference information.

5. [T/F] True or False: a person who gains unauthorized access to computer systems is compiler.

Answer: False

Explanation: Correct answer: hacker/attacker. The source topic discusses hackers in the context of unauthorized access.

6. [T/F] True or False: a strong practice for account security is use strong unique passwords and appropriate authentication.

Answer: True

Explanation: Strong unique credentials reduce account compromise risk.

1. [VSA] Name the concept associated with: malware that disguises itself as legitimate software.

Answer: Trojan horse

Explanation: A Trojan masquerades as legitimate or useful software.

2. [VSA] Name the concept associated with: software that secretly monitors user activity.

Answer: spyware

Explanation: Spyware collects information without appropriate user awareness/consent.

3. [VSA] Name the concept associated with: software that displays unwanted advertisements.

Answer: adware

Explanation: Adware is associated with unwanted advertising.

4. [VSA] Name the concept associated with: malware that encrypts data and demands payment.

Answer: ransomware

Explanation: Ransomware commonly denies access to data and demands a ransom.

5. [VSA] Name the concept associated with: software used to detect and remove known malicious software.

Answer: antivirus

Explanation: Antivirus software detects, blocks, quarantines or removes malware.

6. [VSA] Name the concept associated with: a security system controlling incoming and outgoing network traffic.

Answer: firewall

Explanation: A firewall applies traffic-control rules.

1. [Short/Application] Explain the following concept in detail: software that displays unwanted advertisements.

Answer: adware

Explanation: Adware is associated with unwanted advertising.

2. [Short/Application] Explain the following concept in detail: malware that encrypts data and demands payment.

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Answer: firewall

Explanation: A firewall applies traffic-control rules.

5. [Short/Application] Explain the following concept in detail: the process of converting readable plaintext to ciphertext.

Answer: encryption

Explanation: Encryption transforms plaintext into protected ciphertext.

6. [Short/Application] Explain the following concept in detail: the process of converting ciphertext back to plaintext.

Answer: decryption

Explanation: Decryption reverses encryption when the required key/process is available.

7. [Short/Application] Explain the following concept in detail: a fraudulent attempt to obtain sensitive information by pretending to be trustworthy.

Answer: phishing

Explanation: Phishing uses deceptive messages or sites to obtain information.

